AKADEMIKA

DCL-GMSK

GFSK / GMSK MODULATOR
&
DEMODULATOR KIT
EXPERIMENTAL MANUAL

…. A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO:

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and patch chords are connected correctly as per experiment.

In DCL-GMSK kit, signals are to be monitored with an oscilloscope or with help of indicators as explained in the manual.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity**.**

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

DCL-GMSK: GFSK/GMSK MODULATOR/DEMODULATOR KIT.

DATA SIMULATOR **:** Onboard 8-bit variable NRZ-L pattern.

CRYSTAL OSCILLATOR **:** 32.768 MHz

DATA CLOCK **:** 256 KHz

CARRIER MODULATION **:** GFSK, GMSK.

CARRIER DEMODULATION **:** GFSK, GMSK.

INTERMEDIATE SIGNAL **:** Provision for observing intermediate signals.

INTER CONNECTION **:** 2 mm banana socket.

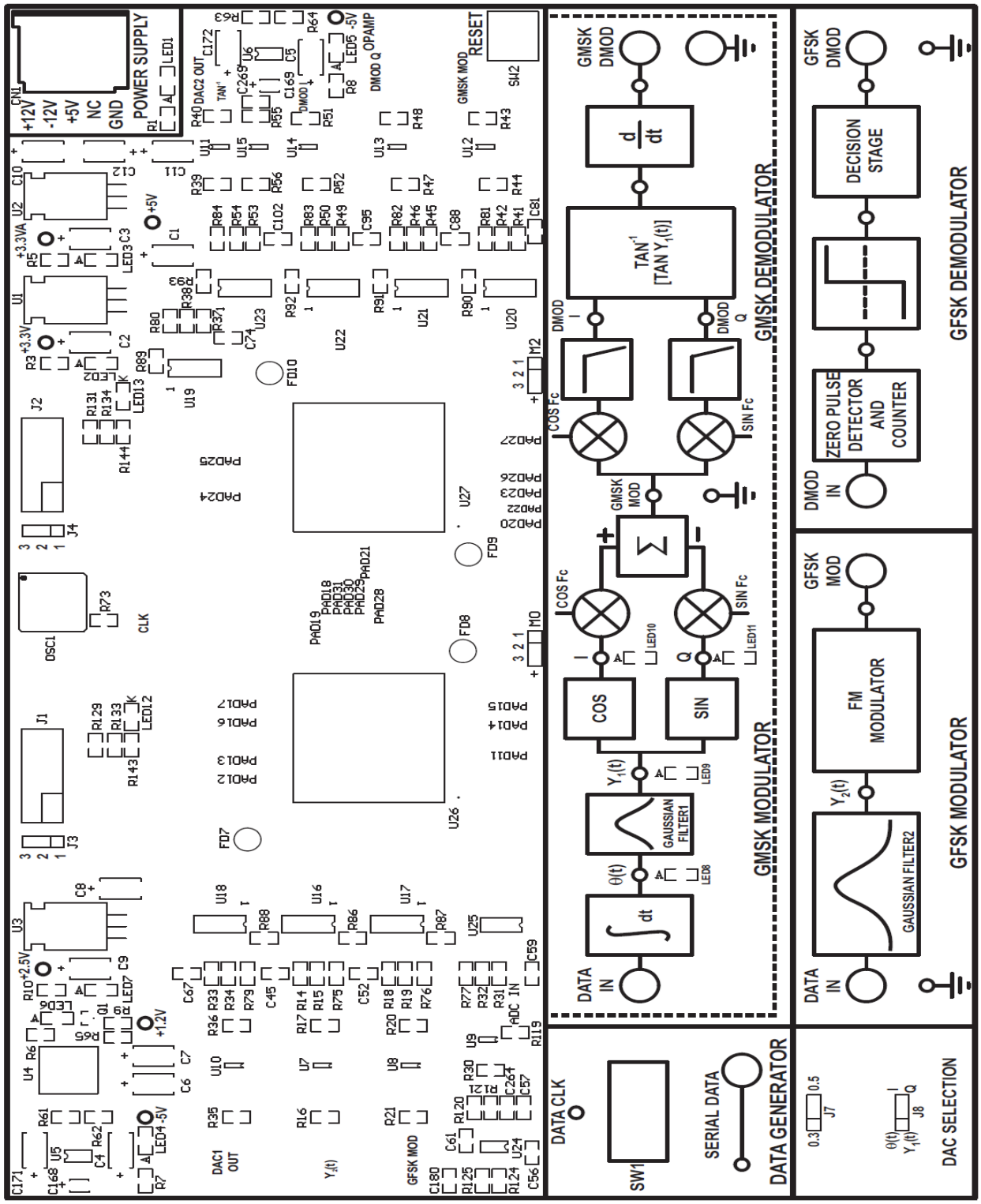
POWER SUPPLY **:** +12V, -12V, +5V, GND.

TEST POINTS **:** 26

ACCESSORIES

	DCL-GMSK	Quantity
1	Short Links PCS2 Patch cord Male To Male	10
2	ADCL-03 EXPT Manual	1
3	5 PIN DIN CONNECTOR Power Supply	1
4	Power cord	1

FUNCTIONAL BLOCK



INTRODUCTION

AKADEMIKA with the trainer, namely, “DCL-GMSK initiates the user to GMSK / GFSK MODULATION AND DEMODULATION technique adopted in practice.

“DCL-GMSK”: GFSK / GMSK MODULATOR / DEMODULATOR KIT.

FEATURES:

1) The trainer comes with an onboard data simulator, which generates the NRZ-L pattern depending on positions of 8-bit switch and data clock of 256 KHz which enables the trainers to work in a stand-alone mode.

2) Two carrier modulation options are available on the kit.

   a) Gaussian Minimum Shift Keying. (GMSK).
   b) Gaussian Frequency Shift Keying. (GFSK).

3) Following options are available for detecting the information from the carrier:

   1) “DMOD I” and “DMOD Q” signals which are the output of the LPF of GMSK Demodulator.
   2) GFSK demodulation using Zero crossing detection techniques.

4) Interconnection facilities:

   Sockets and patch chords are provided for on board connections.

5) Test Points:
   All relevant test points are brought-out for observations. Observations are carried out on DSO.

6) Power Supply:
   +5V, +12V,-12V, GND.

DIGITAL COMMUNICATION:

Throughout years analog modulation techniques were used and are still being used to transmit voice, image and other kinds of information signals. The development of digital techniques, the need for efficiency and demand for higher data rates, made the digital modulation much more popular.

Digital communication systems are becoming increasingly attractive because of the ever growing demand for data communication and because digital transmission offers data processing options and flexibilities not available with analog transmission.

One of the main advantages in the digital communications when compared to analog is the ease of regeneration of the digital signals. The digital circuitry is less susceptible to distortion and interference.

Digital circuits are less subjective to distortion and interference than analog circuits. Because binary digital circuits operate in one or two states-fully on or fully off-to be meaningful, a disturbance must be large enough to change the circuit operating point from one state to other. Such two-state operation facilitates signal regeneration and thus prevents noise and other disturbances from accumulating in transmission.

Analog signals, however, are not two-state signals; they can take an infinite variety of shapes. With analog circuits, even a small disturbance can render the reproduced waveform unacceptably distorted. Once the analog signal is distorted, the distortion cannot be removed by amplification. Because accumulated noise is irrevocably bound to analog signals, they cannot be perfectly regenerated.

With digital techniques, extremely low error rates producing high signal fidelity are possible through error detection and correction but similar procedures are not available with analog.

Thus the move to digital modulation provides more information capacity, compatibility with digital data services, higher data security, and better quality in communication.

Before proceeding further let us discuss about the basic digital modulation techniques and the terminologies used.

MODULATION:

Modulation means to vary or change. The information bearing signals modulate a signal called carrier in a manner that we call modulation, in order to transmit the information through a media either wireless or wired. The modulation process can be done according to any reliable detectable change in signal characteristics. It can be either amplitude or frequency or the phase of the carrier signal that changes according to the information signal. At the receiver, demodulation occurs by detection of changes in the carrier signal.

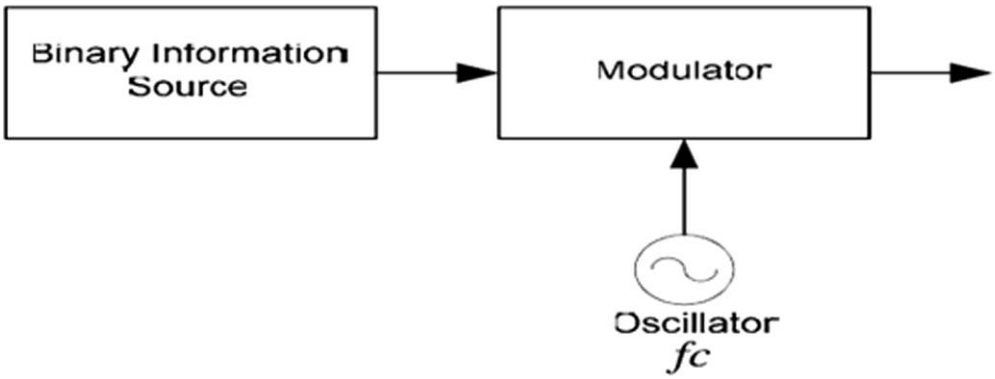


Fig. Basic Transmitter

BASIC DIGITAL MODULATION TECHNIQUES:

1. Amplitude Shift Keying:-

In ASK, the modulator puts out a carrier of specific amplitude for every symbol 1, and no signal for every symbol 0. This scheme is also called on-off keying (OOK). In a general ASK scheme, the amplitude for symbol 0 is not necessarily 0.

2. Frequency Shift Keying:-

Frequency Shift Keying is one of the most popular modulation schemes used widely for Digital Communications. In this scheme, the frequency of the carrier is changed as a function of the modulating signal (data) is being transmitted. Amplitude remains unchanged. In binary FSK (BFSK or 2FSK), a “1” is represented by one frequency and a “0” is represented by another frequency. A particularly useful type of FSK is the CPFSK (Continuous Phase FSK) in which the phase continuity is maintained during the frequency shift. In this modulation scheme because of the phase continuity, a coherent demodulator (Phase Locked Loop) can be used for demodulation.

3. Phase Shift Keying:-

In PSK, the data is contained in the phase of the carrier. In BPSK (Binary PSK) a phase shift of 0 or 360 degrees signifies a logic zero and a phase shift of 180 degrees signifies a logic one. In QPSK (Quadrature PSK), which is treated as a more bandwidth efficient scheme, 4 phase shifts of 0^0, 90^0, 180^0, 270^0 indicate 4 different levels. Thus providing double data rate in the same bandwidth of BPSK. But because of the greater phase shifts, QPSK- envelope is not constant as we wish it to be. It goes to zero once every 180 or 270 degree transition. So OQPSK (Offset QPSK) --- a variant is used, in which the two bits that are parallel encoded are slightly offsetted so that at any instant the maximum phase shift is 90 degrees. But still the transitions are not as smooth as we want them to be.

To conclude with fig 2 shows short summary of three basic binary carrier modulations. They are amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK).

In ASK the modulator puts out a burst of carrier for every symbol 1, and no signal for every symbol 0.

In FSK, for symbol 1 a higher frequency burst is transmitted and for symbol 0 a lower frequency burst is transmitted, or vice versa.

In PSK, a symbol 1 is transmitted as a burst of carrier with 0 initial phase while a symbol 0 is transmitted as a burst of carrier with 180◦ initial phase.

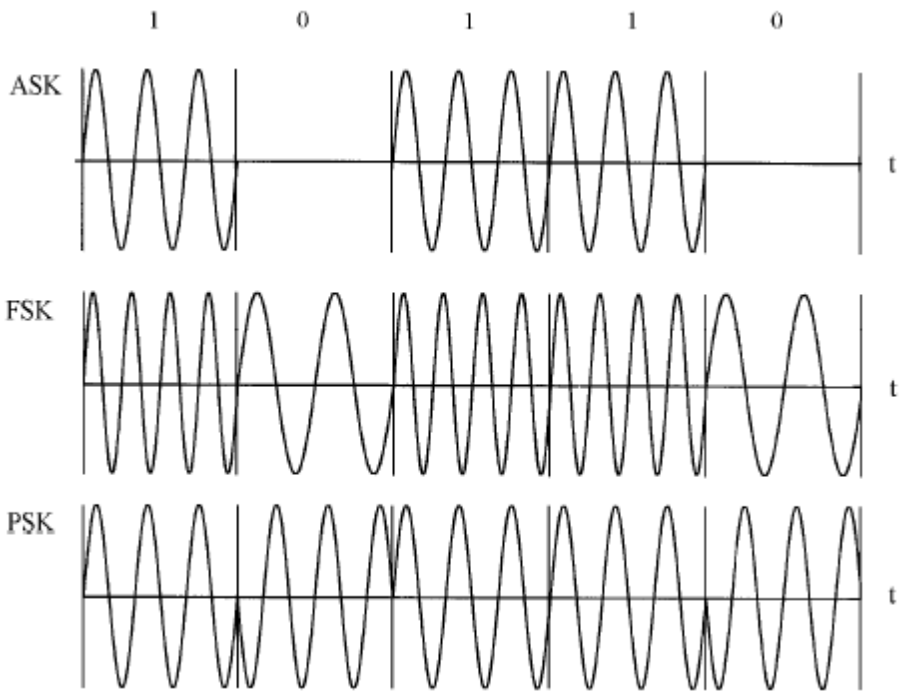


Fig 2. THREE BASIC MODULATION SCHEME

Based on these three basic schemes, a variety of modulation schemes can be derived from their combinations. For example, by combining FSK and PSK signals with orthogonal carriers a new scheme called Minimum Shift Keying (MSK) can be generated. GMSK is the derived form of MSK and GFSK is the derived form of FSK.

CRITERIA OF CHOOSING MODULATION SCHEMES:

The essence of digital modem design is to efficiently transmit digital bits and recover them from corruptions from the noise and other channel impairments. There are three primary criteria of choosing modulation schemes: power efficiency, bandwidth efficiency, and system complexity.

- **POWER EFFICIENCY:**

Power efficiency describes the ability of the system to reliably send information at the lowest practical power level.

- **BANDWIDTH EFFICIENCY:**

Bandwidth efficiency describes the ability of a modulation scheme to accommodate data within a limited bandwidth.

- **SYSTEM COMPLEXITY:**

System complexity refers to the amount of circuits involved and the technical difficulty of the system. Associated with the system complexity is the cost of manufacturing, which is of course a major concern in choosing a modulation technique. Usually the demodulator is more complex than the modulator. Coherent demodulator is much more complex than non-coherent demodulator since carrier recovery is required.

Before proceeding further let us discuss about the concept of Inter-symbol Interference (ISI).

Inter-symbol Interference (ISI) is a situation in which the energy from one symbol slot is spread out over neighboring symbol slots. ISI may be introduced either by the channel, when the RMS delay spread becomes an appreciable fraction of the bit period, or by the filtering of the data pulses, in order to reduce the out of band power, before the modulation process.

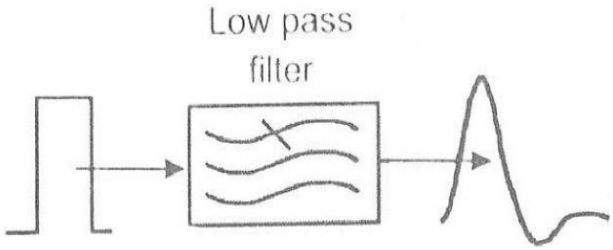


Fig 3. Filtering Effect of the Channel

When a sequence of signals spread, parts of the symbol energy overlap with neighboring symbols causing inter-symbol interference (ISI).

Minimum Shift Keying:-

The motivation behind MPSK was to increase the bandwidth efficiency of the PSK modulation schemes.

In BPSK, a data bit is represented by a symbol. In MPSK, n = log2 M data bits are represented by a symbol, thus the bandwidth efficiency is increased to n times. Among all MPSK schemes, QPSK is the most-often-used scheme since it does not suffer from BER degradation while the bandwidth efficiency is increased. Other MPSK schemes increase bandwidth efficiency at the expense of BER performance.

The major advantage of OQPSK over QPSK is that it exhibits less phase changes at symbol transitions, thus out-of-band interference due to band limiting and amplifier nonlinearity is reduced.

This suggests that further improvement is possible if phase transitions are further smoothed or even become completely continuous. Minimum shift keying (MSK) is such a continuous phase modulation scheme. It can be derived from OQPSK by shaping the pulses with half sinusoidal waveforms, or can be derived as a special case of continuous phase frequency shift keying (CPFSK).

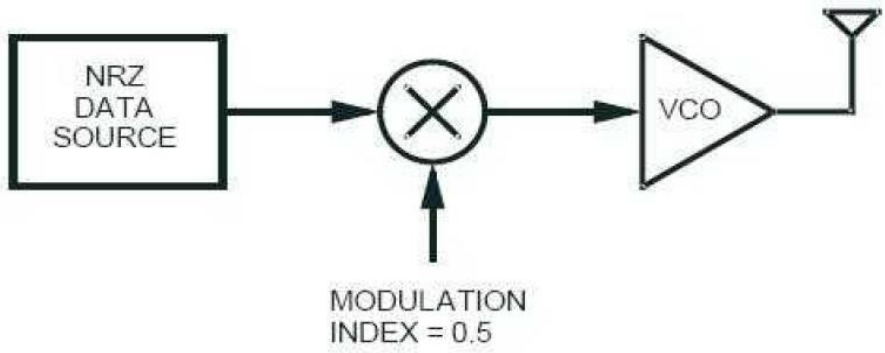
Minimum Shift Keying can be viewed in either of the following ways:-
1. Variant of CPFSK
2. Variant of OQPSK

Minimum Shift Keying is CPFSK such that the maximum frequency deviation is ½ of the data rate. That is the Mark and Space frequencies are spaced ½ of Data Rate apart. This is the minimum spacing with which a coherent demodulator can decode the FSK properly. Because it occupies the minimum Bandwidth among any FSK variant it is referred to as Minimum Shift Keying.
In other words, the modulation index m is 0.5 for MSK, and is defined as

$$m = \Delta f \times T$$

where, $\Delta f = f_{mark} - f_{space}$

$T = 1/\text{bit rate}$



Direct MSK modulation

Alternatively, MSK can be viewed as OQPSK with the bits shaped as half sinusoids rather than pulses. Such Pulse shaping reduces the bandwidth required, thereby making it more Bandwidth Efficient than OQPSK.

MSK has a narrower spectrum than wider deviation forms of FSK. The width of the spectrum is also influenced by the waveforms causing the frequency shift. If those waveforms have fast transitions or a high slew rate, then the spectrum of the transmitter will be broad.

Even MSK was not sufficient to satisfy the stringent Bandwidth criterion of the Mobile environment. The out-of-Band energy in MSK was still not low enough to prevent out-of-band interference. To overcome this problem GMSK and GFSK modulation techniques are used in which rectangular pulses are shaped by a Gaussian Filter before doing the modulation. The LPF was chosen so that the out-of-band power is minimized.
Before going to study the Gaussian Filter first we study the concept of Pulse Shaping.

PULSE SHAPING:

Need of Pulse Shaping:

Transmitting a signal at high modulation rate through a band-limited channel can create inter symbol interference. As the modulation rate increases, the signal's bandwidth increases. When the signal's bandwidth becomes larger than the channel bandwidth, the channel starts to introduce distortion to the signal. This distortion is usually seen as inter symbol interference.

The signal's spectrum is determined by the pulse shaping filter used by the transmitter. Usually the transmitted symbols are represented as a time sequence. This theoretical signal is then filtered with the pulse shaping filter, producing the transmitted signal. The spectrum of the transmission is thus determined by the filter.

In other words, the channel for the signal is band-limited. Therefore better filters have been developed, which attempt to minimize the bandwidth needed for a certain symbol rate.

Some common pulse shaping methods are

- Root Raised Cosine (use with QPSK)
- Half-Sinusoidal (use with MSK)
- Gaussian (used with GMSK and GFSK)
- Quadrature Partial Response (QPR)

In general, the MPSK (M'ary phase shift keying) spectrum consists of a main lobe representing the middle of the spectrum and various side lobes located on either side of the main lobe. Shaping the spectrum should satisfy two criteria: The main lobe should be as narrow as possible, and the maximum side lobe level should be as small as possible relative to the main lobe.

GAUSSIAN FILTER:

A **Gaussian filter** is a filter whose impulse response is a Gaussian function. Gaussian filters are designed to give no overshoot to a step function input. This behavior is closely connected to the fact that the Gaussian filter has the minimum possible group delay. Mathematically, a Gaussian filter modifies the input signal by convolution with a Gaussian function.

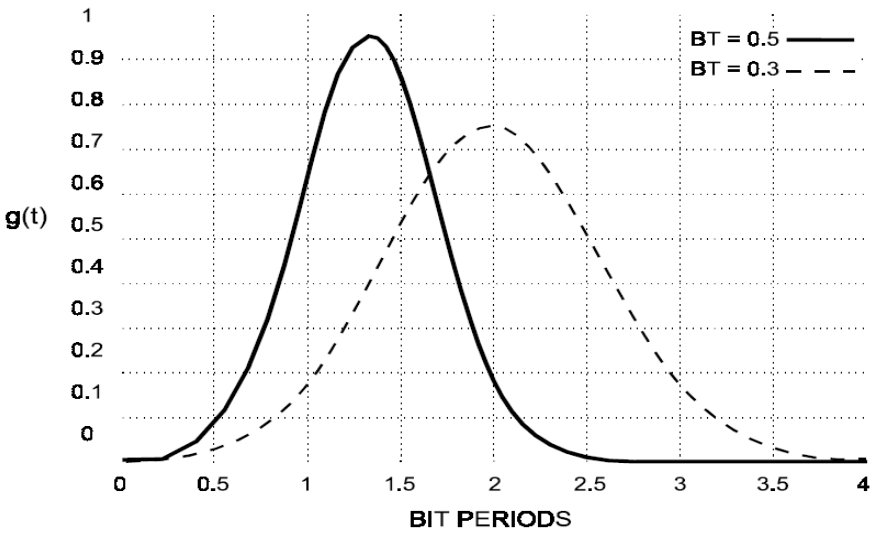


Fig. Gaussian Filter Impulse Response for BT = 0.5 and BT = 0.3

- Characteristics of Gaussian Filter
  - Narrow bandwidth and sharp cutoff to suppress high frequency components.
  - Small overshoot impulse response to prevent excess deviation of the instantaneous frequency.
- Impulse Response of Gaussian Filter
  - Impulse response defined by a Gaussian Distribution – no overshoot or ringing.
  - The impulse response of the Gaussian Filter is:

$$h(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-t^2}{2\sigma^2}\right)$$

Where б = standard deviation

$$\sigma = \frac{\sqrt{\ln(2)}}{2\pi BT}$$

- BT – refers to the filter's -3dB BW and data rate by:

$$BT = \frac{f_{-3dB}}{BitRate}$$

- B is the -3 dB bandwidth of the filter. The square pulse response g(t) of the Gaussian LPF is:

$$g(t) = h(t) * \mathrm{rect}\left(\frac{t}{T}\right)$$

where the rectangular function rect(x) is defined by:

$$\mathrm{rect}\left(\frac{t}{T}\right) = \begin{cases} 1/T, & \text{for } |t| < \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$

- For GMSK BT = 0.3 and for GFSK BT = 0.5.
- For BT=0.3, adjacent symbols will interfere with each other more than for BT=0.5
- GMSK with BT= ∞ is equivalent to MSK.

- Applications of Gaussian Filter
  - Gaussian pulse shaping of the baseband pulse stream is resorted to in digital communication systems like DECT and GSM in order to limit the spectral energy outside the transmission band.

In today's communication systems, the channel separations and the channel bandwidths are restricting tasks. In order to use the frequency spectrum efficiently, standards have been developed for different communication systems. To use the communication channel efficiently, without drawback from the information capacity, new modulation schemes like GMSK, GFSK have been developed and use.

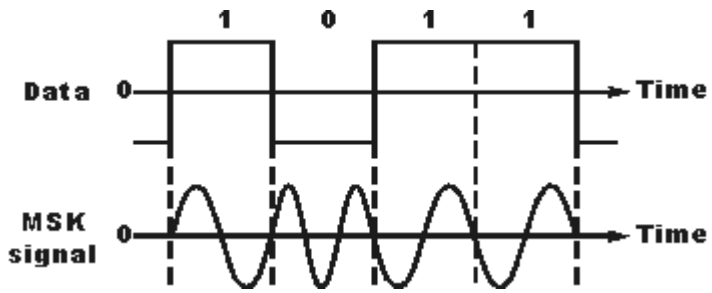
Gaussian Minimum Shift Keying (GMSK)

GMSK basics

GMSK modulation is based on MSK, which is itself a form of continuous-phase frequency-shift keying. One of the problems with standard forms of PSK is that sidebands extend out from the carrier. To overcome this, MSK and its derivative GMSK can be used.

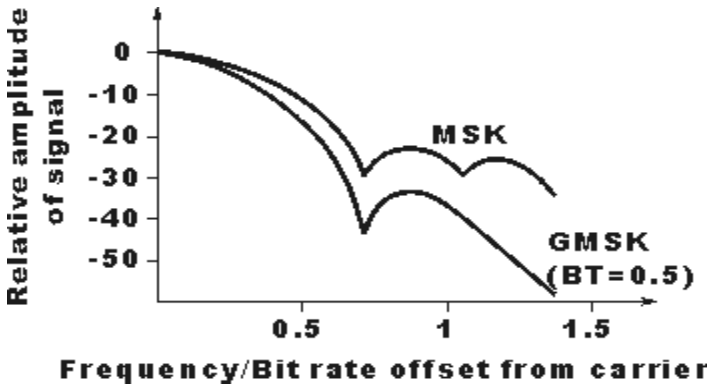
MSK and also GMSK modulation are what is known as a continuous phase scheme. Here there are no phase discontinuities because the frequency changes occur at the carrier

zero crossing points. This arises as a result of the unique factor of MSK that the frequency difference between the logical one and logical zero states is always equal to half the data rate. This can be expressed in terms of the modulation index, and it is always equal to 0.5.



Signal using MSK modulation

A plot of the spectrum of an MSK signal shows sidebands extending well beyond a bandwidth equal to the data rate. This can be reduced by passing the modulating signal through a low pass filter prior to applying it to the carrier. The requirements for the filter are that it should have a sharp cut-off, narrow bandwidth and its impulse response should show no overshoot. The ideal filter is known as a Gaussian filter which has a Gaussian shaped response to an impulse and no ringing. In this way the basic MSK signal is converted to GMSK modulation.



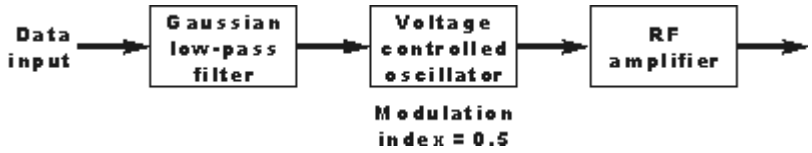
Spectral density of MSK and GMSK signals

The pulse response of Gaussian Low Pass Filter has a direct impact on both spectrum and performance of GMSK system. This response is determined by the BT product where B is the 3 dB bandwidth of the Gaussian Filter and T is the bit duration. BT = ∞ corresponds to MSK system. A smaller BT leads to more compact spectrum but at the same time more ISI. Hence choice of BT is a compromise between spectrum efficiency and BER performance. So taking into consideration the usage of the spectrum and the problem of ISI the choice of BT for GMSK is 0.3.

Generating GMSK modulation

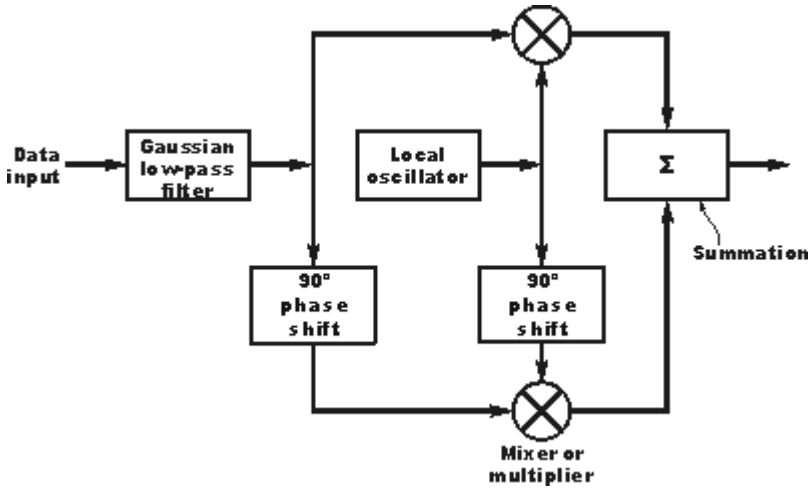
There are two main ways in which GMSK modulation can be generated. The most obvious way is to filter the modulating signal using a Gaussian filter and then apply this to a frequency modulator where the modulation index is set to 0.5. This method is very simple

and straightforward but it has the drawback that the modulation index must exactly equal 0.5. In practice this analogue method is not suitable because component tolerances drift and cannot be set exactly.



Generating GMSK using a Gaussian filter and VCO

A second method is more widely used. Here what is known as a quadrature modulator is used. The term quadrature means that the phase of a signal is in quadrature or 90 degrees to another one. The quadrature modulator uses one signal that is said to be in-phase and another that is in quadrature to this. In view of the in-phase and quadrature elements this type of modulator is often said to be an I-Q modulator. Using this type of modulator the modulation index can be maintained at exactly 0.5 without the need for any settings or adjustments. This makes it much easier to use, and capable of providing the required level of performance without the need for adjustments. For demodulation the technique can be used in reverse.



Block diagram of I-Q modulator used to create GMSK

Advantages of GMSK:-

- Spectral Efficiency
- Constant RF Envelope (Unlike OQPSK)
- Compatible with FM Analog Modems.
- Reasonable Performance in Multi-path Fading Environment

GAUSSIAN FREQUENCY SHIFT KEYING [GFSK]

GFSK can be viewed as a form of the Continuous-Phase Frequency Shift Keying (CPFSK). In CPFSK modulation, the high-frequency components in the output spectrum of the modulated signal are reduced because of the continuous phase variation of the CPFSK modulated signal. In GFSK, the baseband signal is passed through a Pulse-Shaping Gaussian Low Pass Filter before modulation in order to shape the pulses to give them Gaussian shape so that the phase trajectory of the FSK signal becomes smooth and

the instantaneous frequency variations over time are stabilized. This has the following advantages:

1) The envelope of the modulated signal is constant. This allows the GFSK modulated signal to be operated with a Class-C Power Amplifier without introducing Spectrum Regeneration. Therefore, lower power consumption and higher power efficiency can be achieved.

2) The output spectrum has a narrow Main Lobe and a lower level of spectral Side Lobes than in Discontinuous-Phase FSK. This keeps the adjacent channel interference to low levels achieving high spectral efficiency. This is important for a band limited channel and particularly important when the channel is nonlinear.

3) The GFSK modulated signal can be demodulated by Non-Coherent Demodulation schemes leading to low-cost GFSK receivers.

In Gaussian prefiltered FSK systems (GFSK), the modulation index m can be chosen in a wide range in several wireless applications.
And the choice of BT product for GFSK is 0.5.

CIRCUIT DESCRIPTION

- **DATA GENERATOR**

  This block generates NRZ-L data of a variable pattern depending on positions of eight bit switch SW1 also Reference clock of frequency 256KHz (DATA CLK) is generated. The crystal oscillator generates a 32.768MHz clock. The output of this block is serial data and data clock of 256 KHz. The logic for serial data generation and clock generation is written in VHDL.

  256 KHz clock is generated by using 10-bit binary down counter.

  For data generation output of SW1 i.e. parallel data and Load/Shift pulse of 32 KHz is given to Parallel to serial shift register. Output of shift register is taken as serial data.

- **GMSK MODULATOR SECTION**

  Refer sheet 4, 5 and 9 of DCL-GMSK for the Circuit diagram of GMSK MODULATOR. This section consist DAC1, DAC2, DAC GMSK MOD and FPGA. Jumper J8 is provided for the selection of DAC i.e. DAC1 and DAC2. The logic for GMSK MODULATOR is written in the VHDL. The digital inputs to the DAC THS 5641A (U18, U19 and U20) are fed by the FPGA (U27). All DACs works in mode 1 configuration. The final output is given by the IC OPA 690 (U10, U11 and U12) at pin no.1.

- **GMSK DEMODULATOR SECTION**

  Refer sheet 5, 6 and 9 of DCL-GMSK for the Circuit diagram of GMSK DEMODULATOR. This section consist DAC DEMOD I, DAC DMOD Q, DAC ARCTAN and FPGA. Input to the GMSK modulator is fed internally. The logic for GMSK DEMODULATOR is written in the VHDL. The digital inputs to the DAC THS 5641A (U21, U22 and U23) are fed by the FPGA (U27). All DACs works in mode 1 configuration. The final output is given by the IC OPA 690 (U13, U14 and U15) at pin no.1.

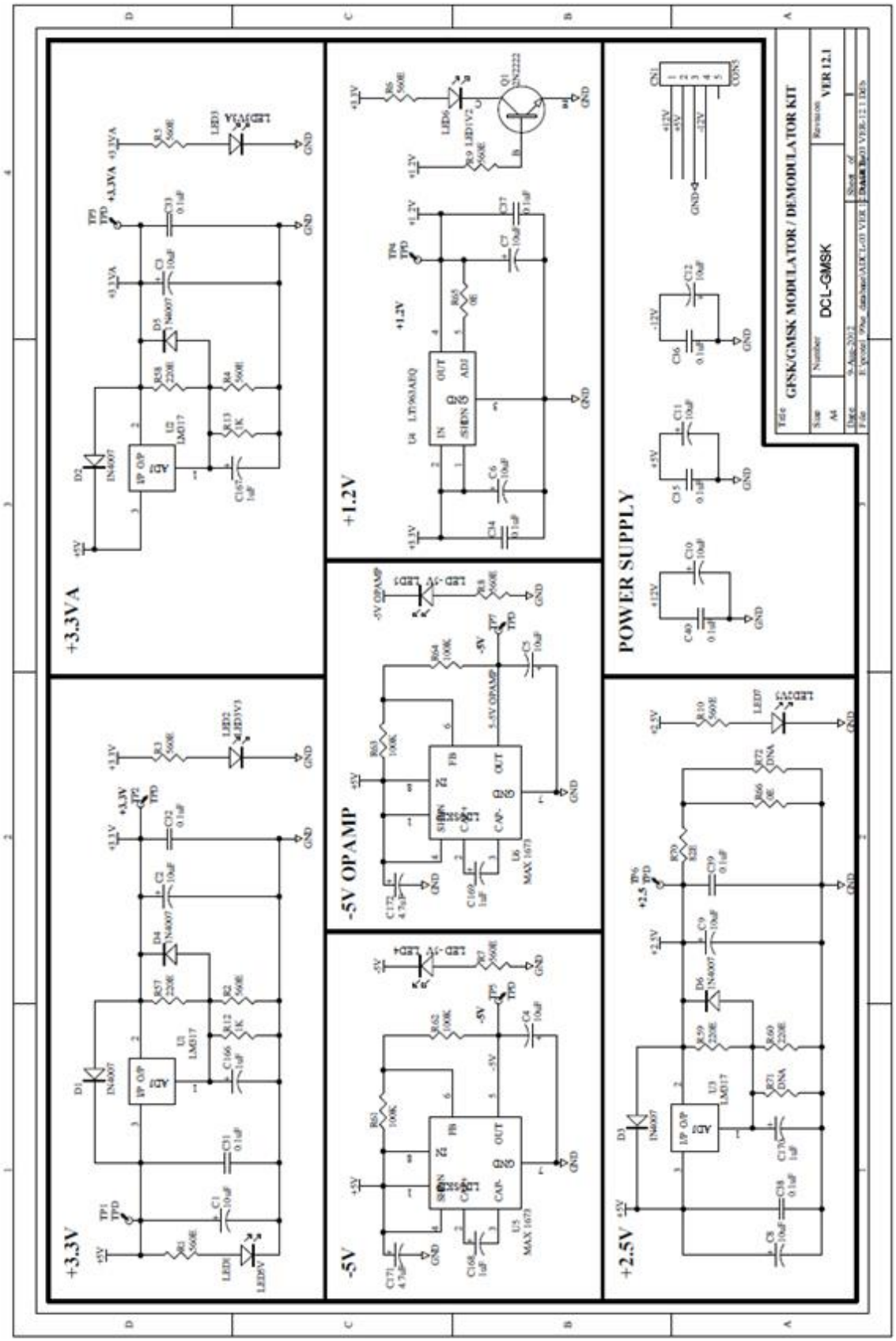
- **GFSK MODULATOR SECTION**

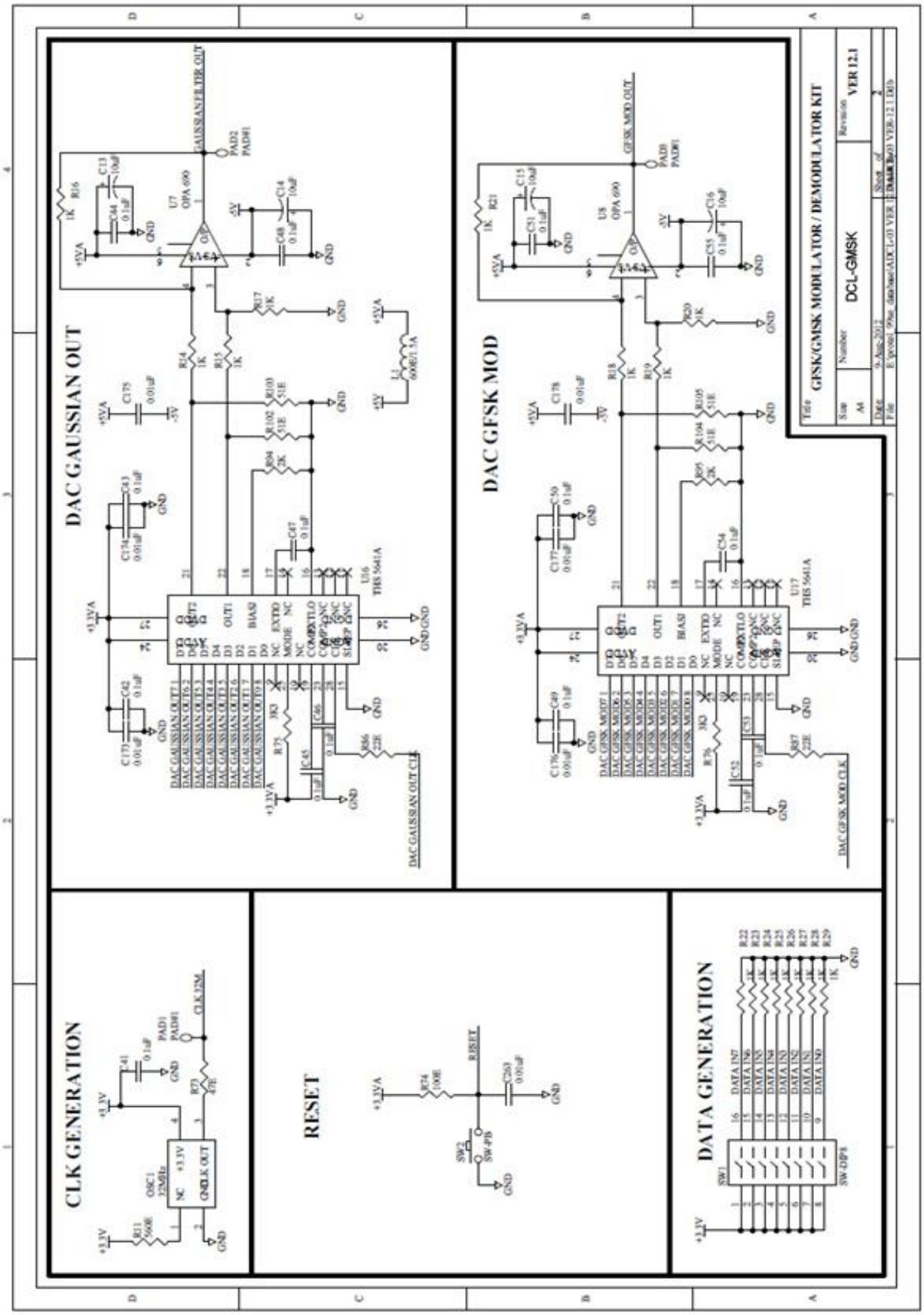
  Refer sheet 4, 5 and 7 of DCL-GMSK for the Circuit diagram of GFSK MODULATOR. This section consist DAC GAUSSIAN, DAC GFSK MOD and FPGA. The logic for GFSK MODULATOR is written in the VHDL. The digital inputs to the DAC THS 5641A (U16 and U17) are fed by the FPGA (U26). All DACs works in mode 1 configuration. The final output is given by the IC OPA 690 (U7 and U8) at pin no.1.

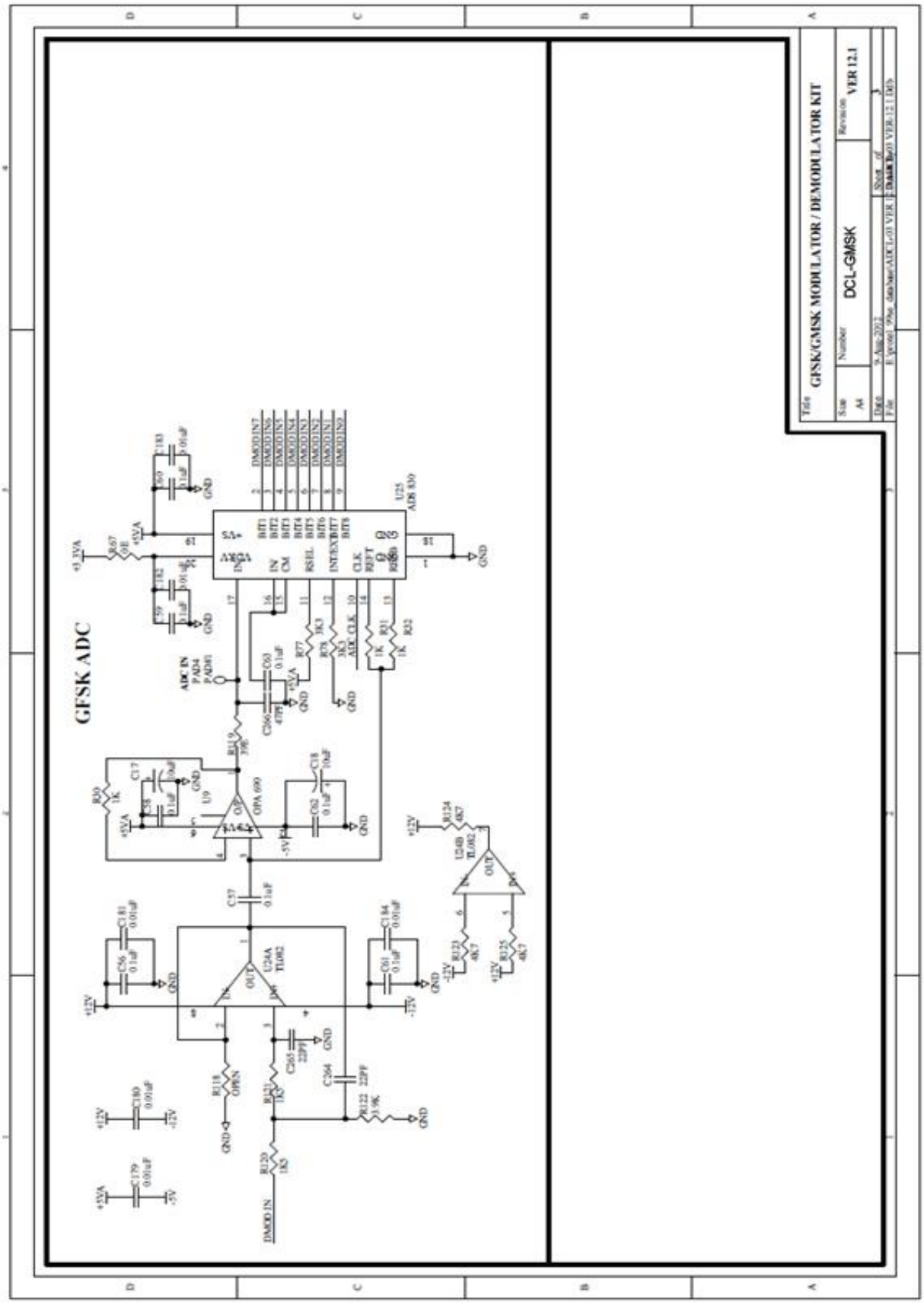
- **GFSK DEMODULATOR SECTION**

  Refer sheet 3 and 7 of DCL-GMSK for the Circuit diagram of GFSK DEMODULATOR. This section consists of IC TL082 (U24) as Low Pass Filter to remove high frequency components from the modulated signal, IC OPA 690 (U9) as DC shifter to give 2V dc shift to the filter output and IC ADS 830 (U25) for A to D Conversion. This digital output of IC U25 is given to the FPGA (U26) for demodulation. The logic for GFSK DEMODULATOR is written in the VHDL.

CIRCUIT DIAGRAM OF DCL-GMSK




DAC GAUSSIAN OUT
DAC GFSK MOD
CLK GENERATION
RESET
DATA GENERATION
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1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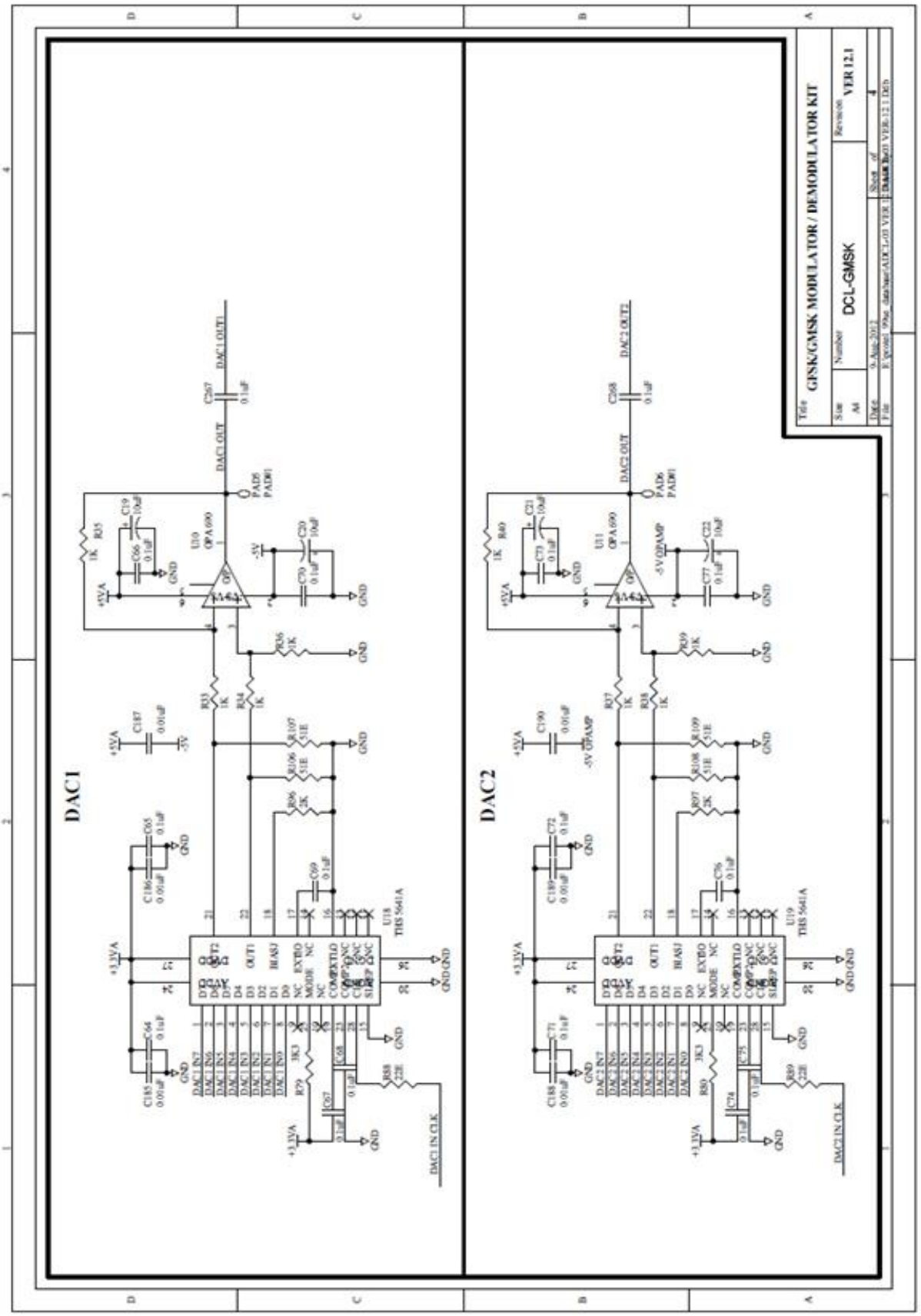
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1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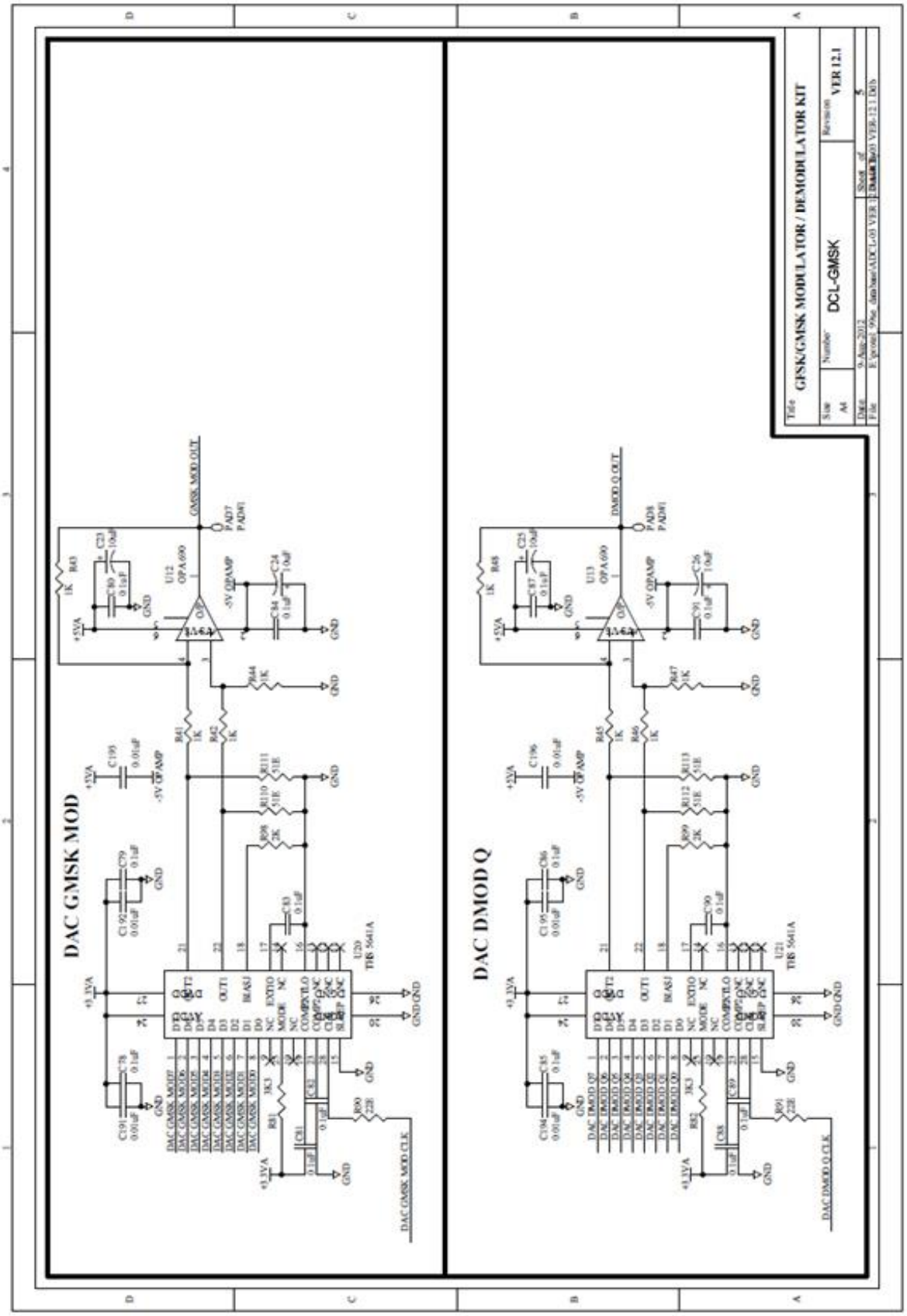
DAC2
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1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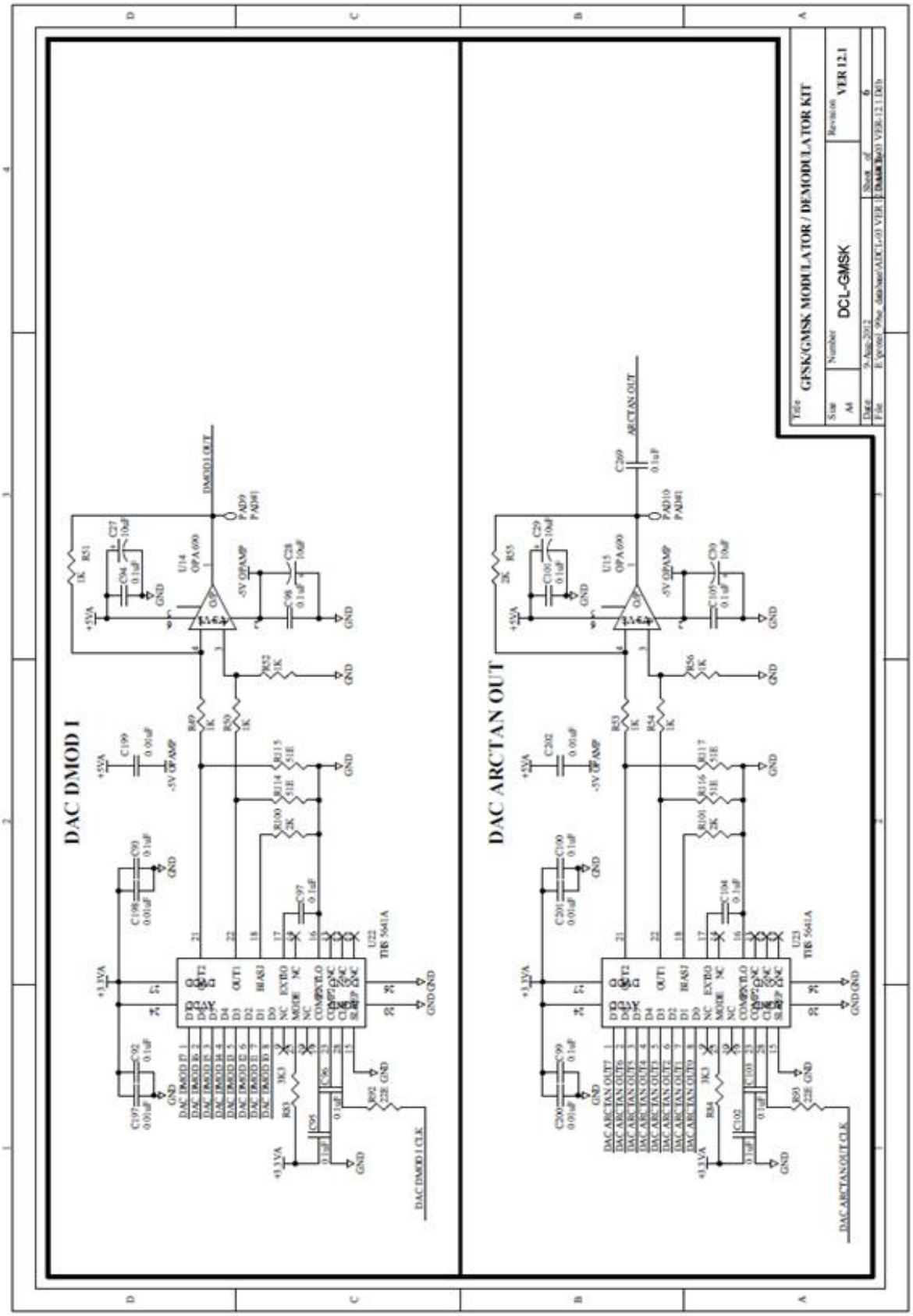
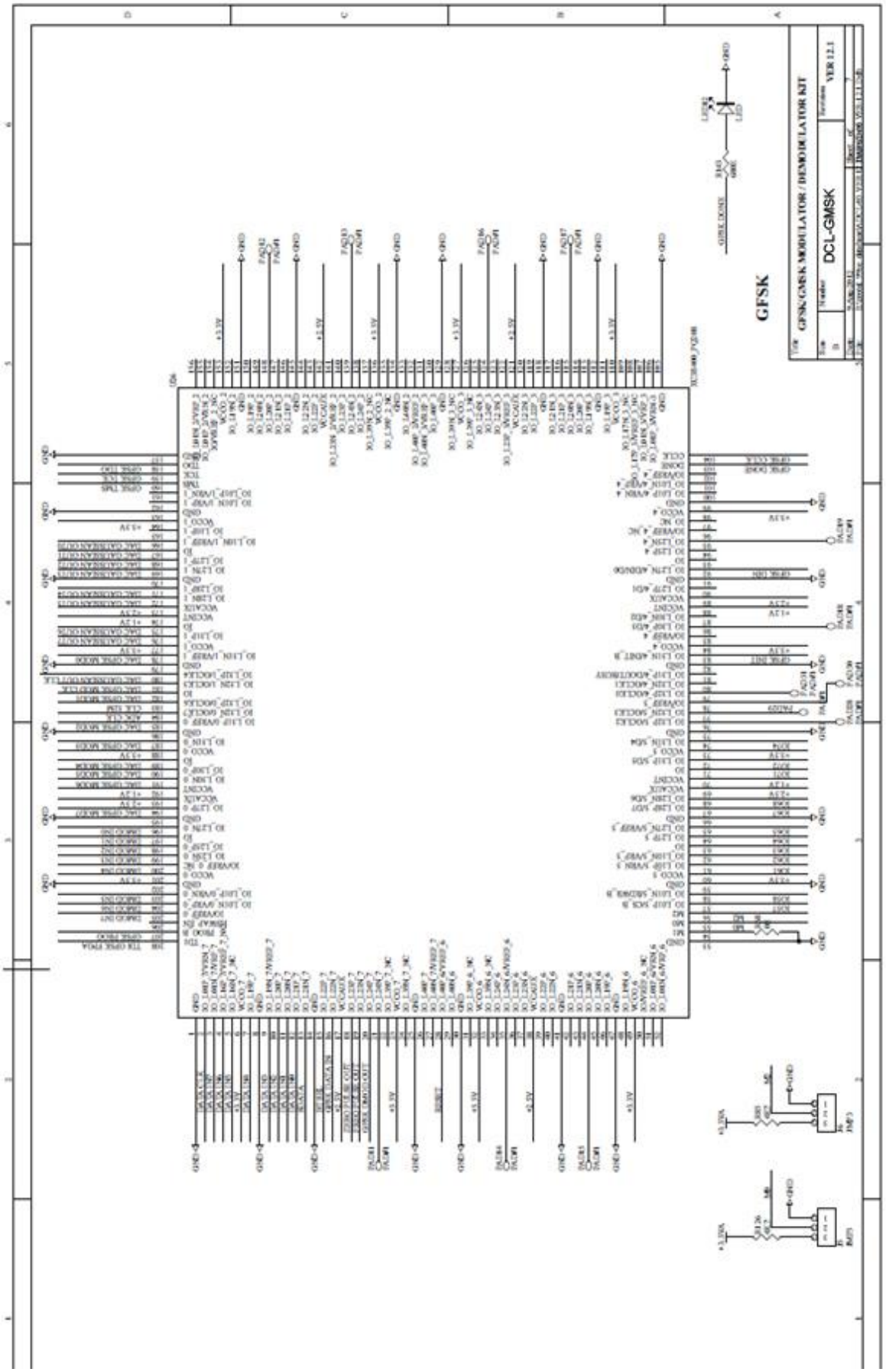
DAC DMOD Q
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1


DAC DMOD I
DAC ARCTAN OUT
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1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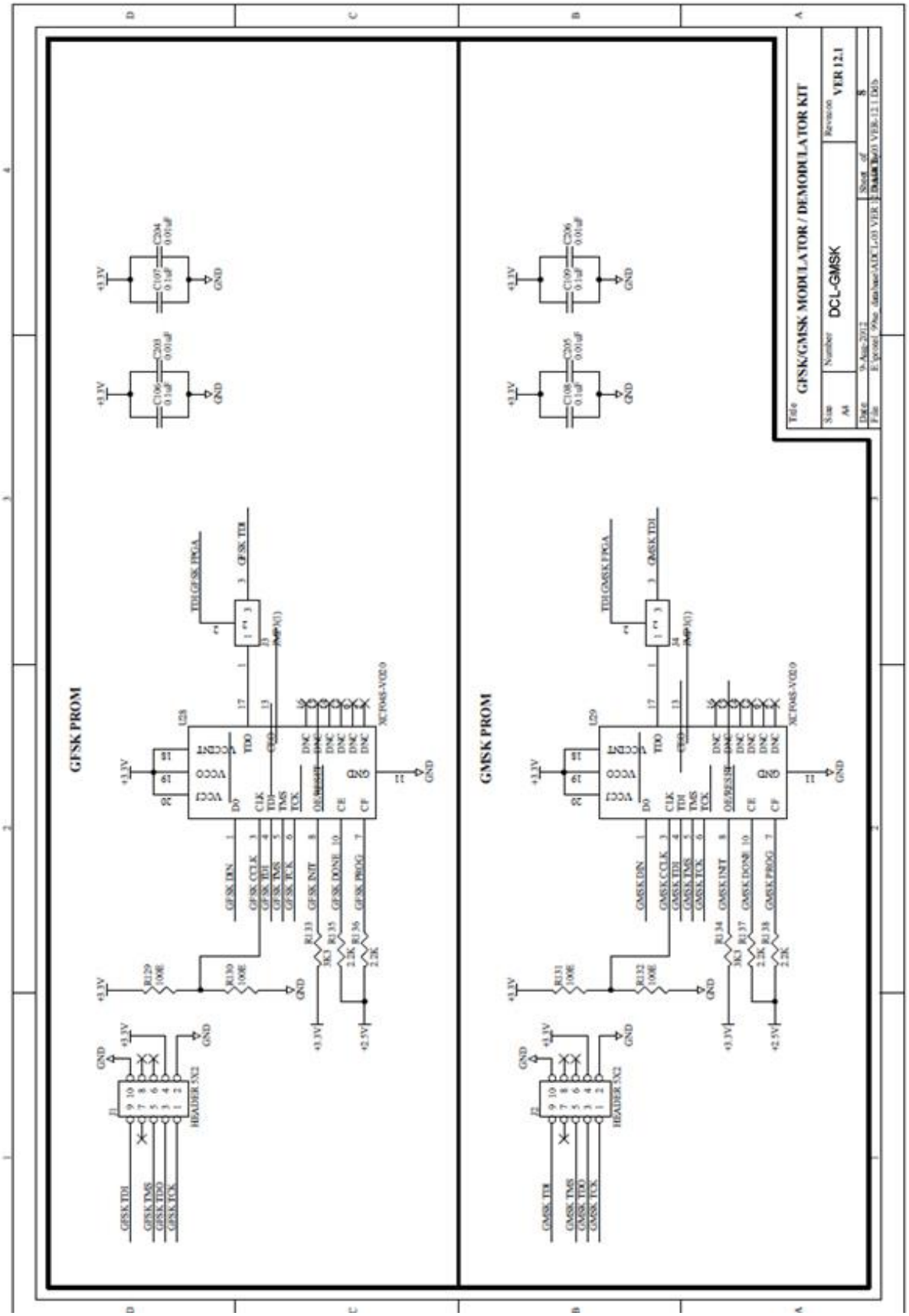
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1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GMSK PROM
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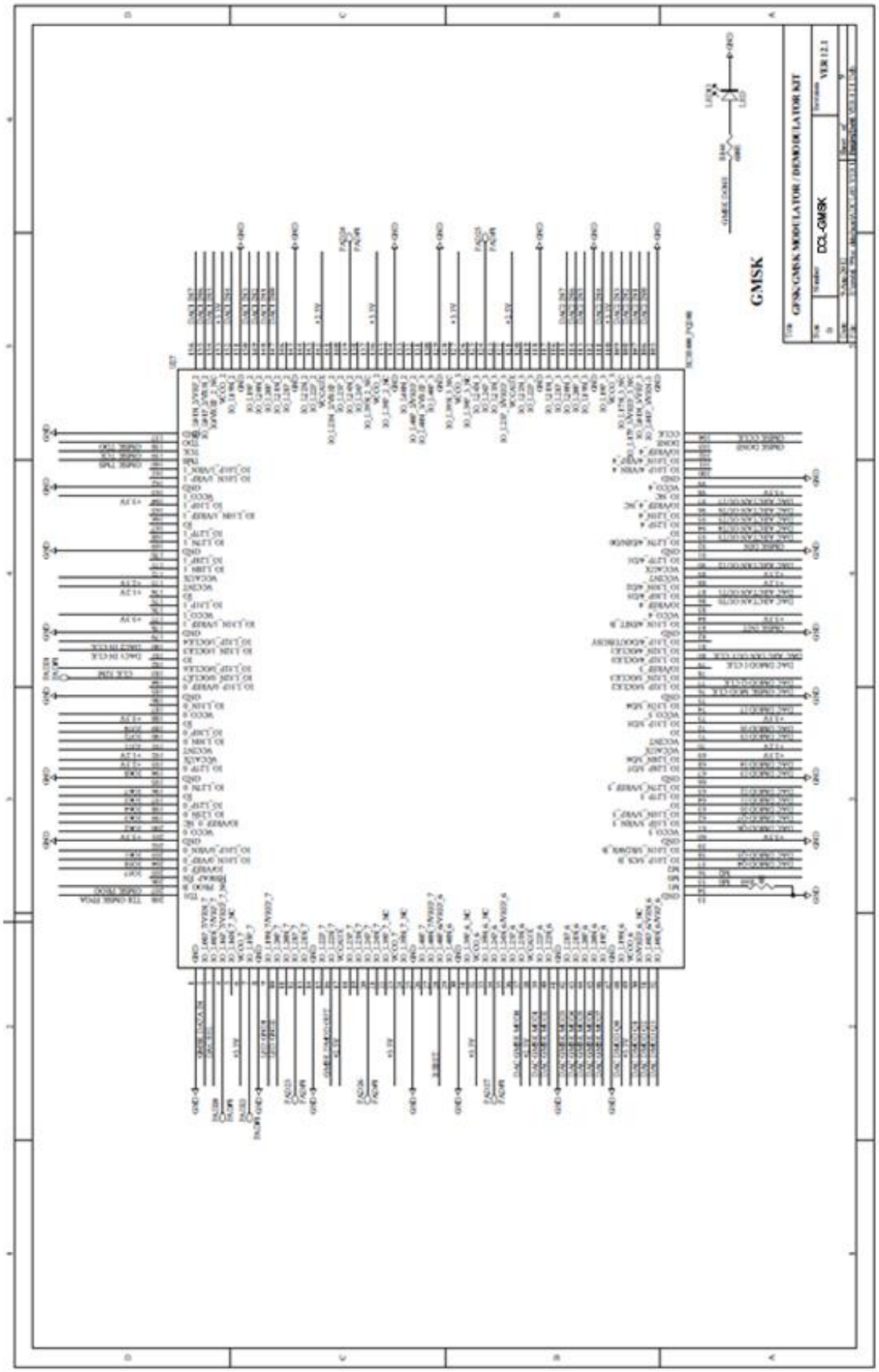
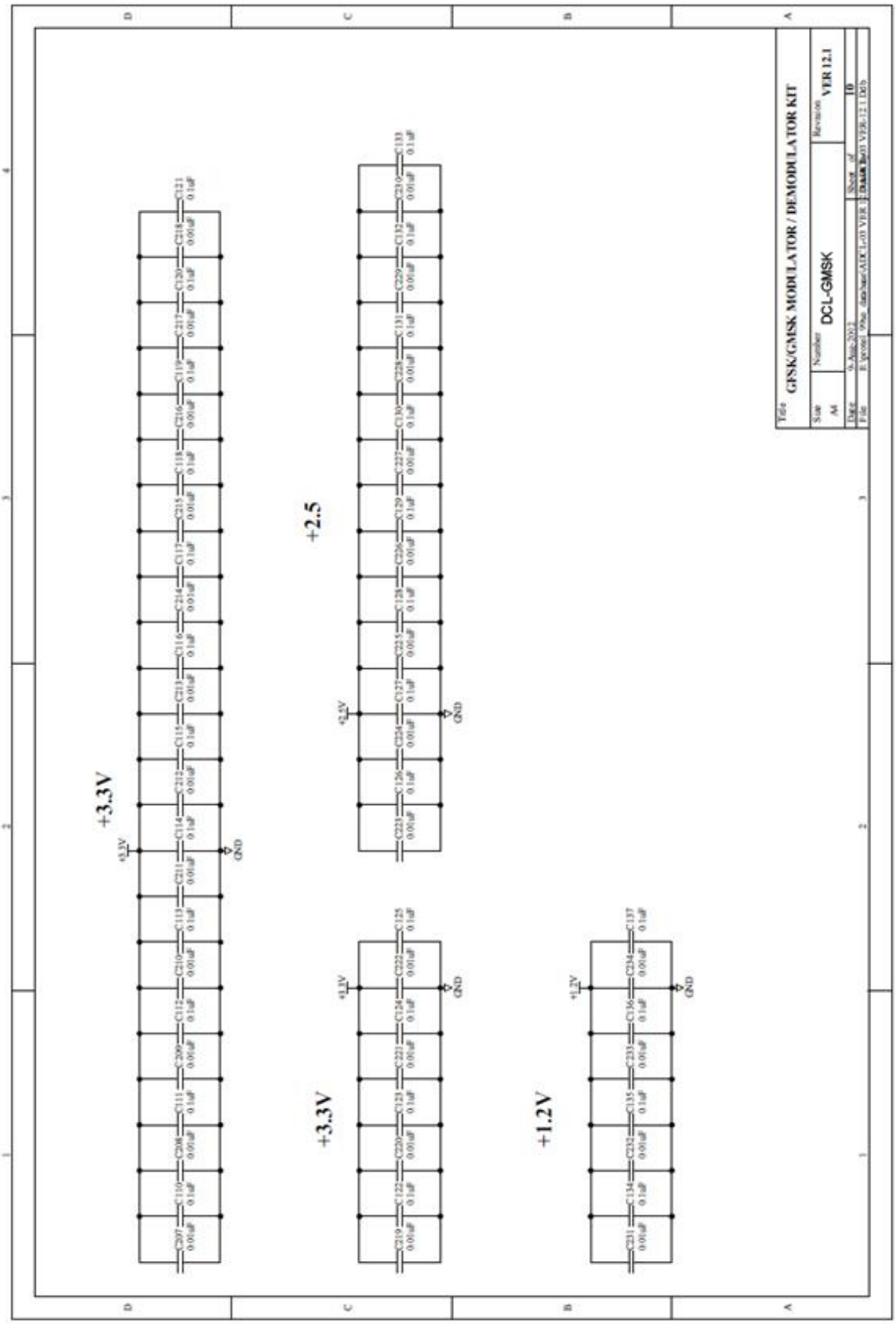
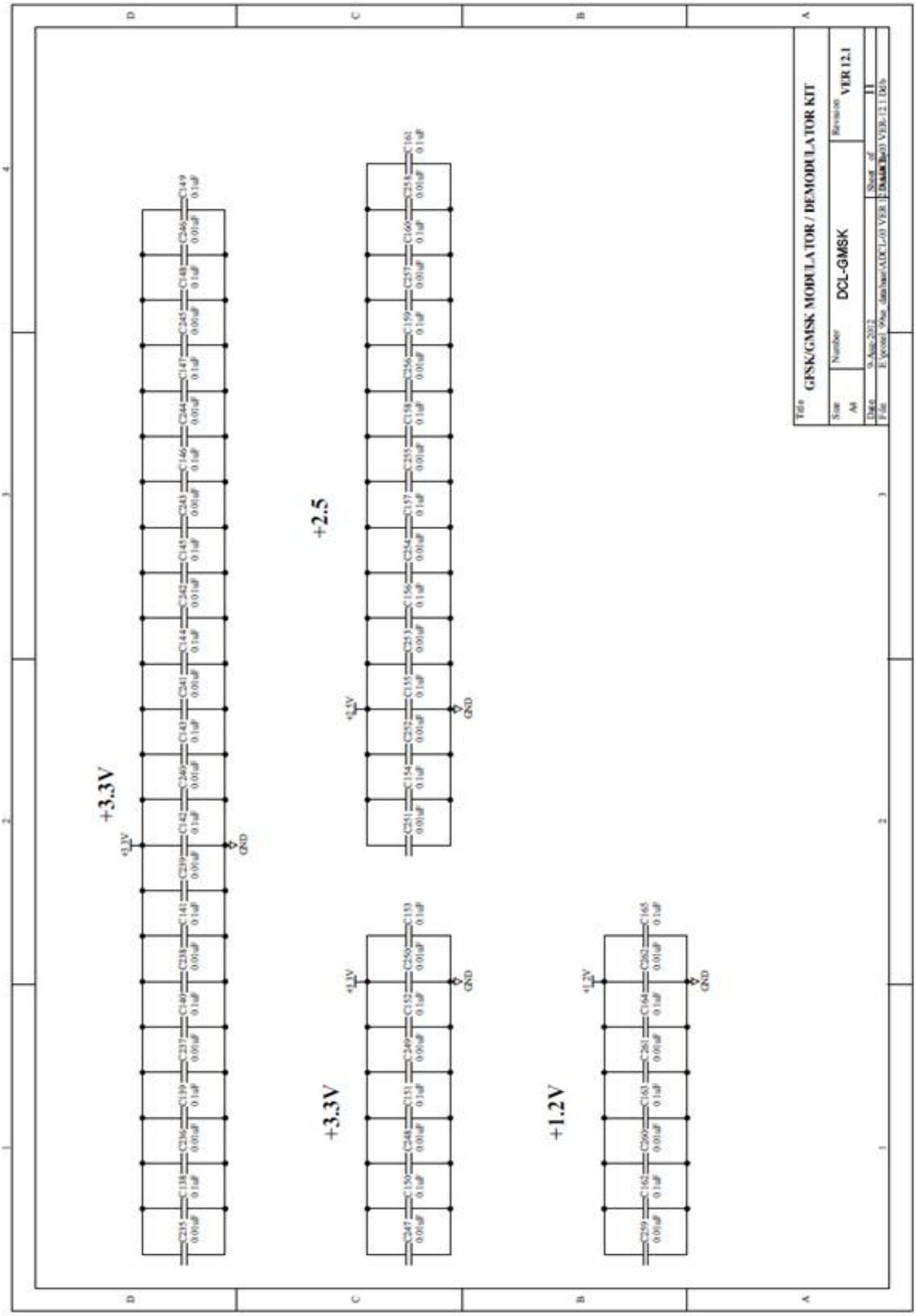
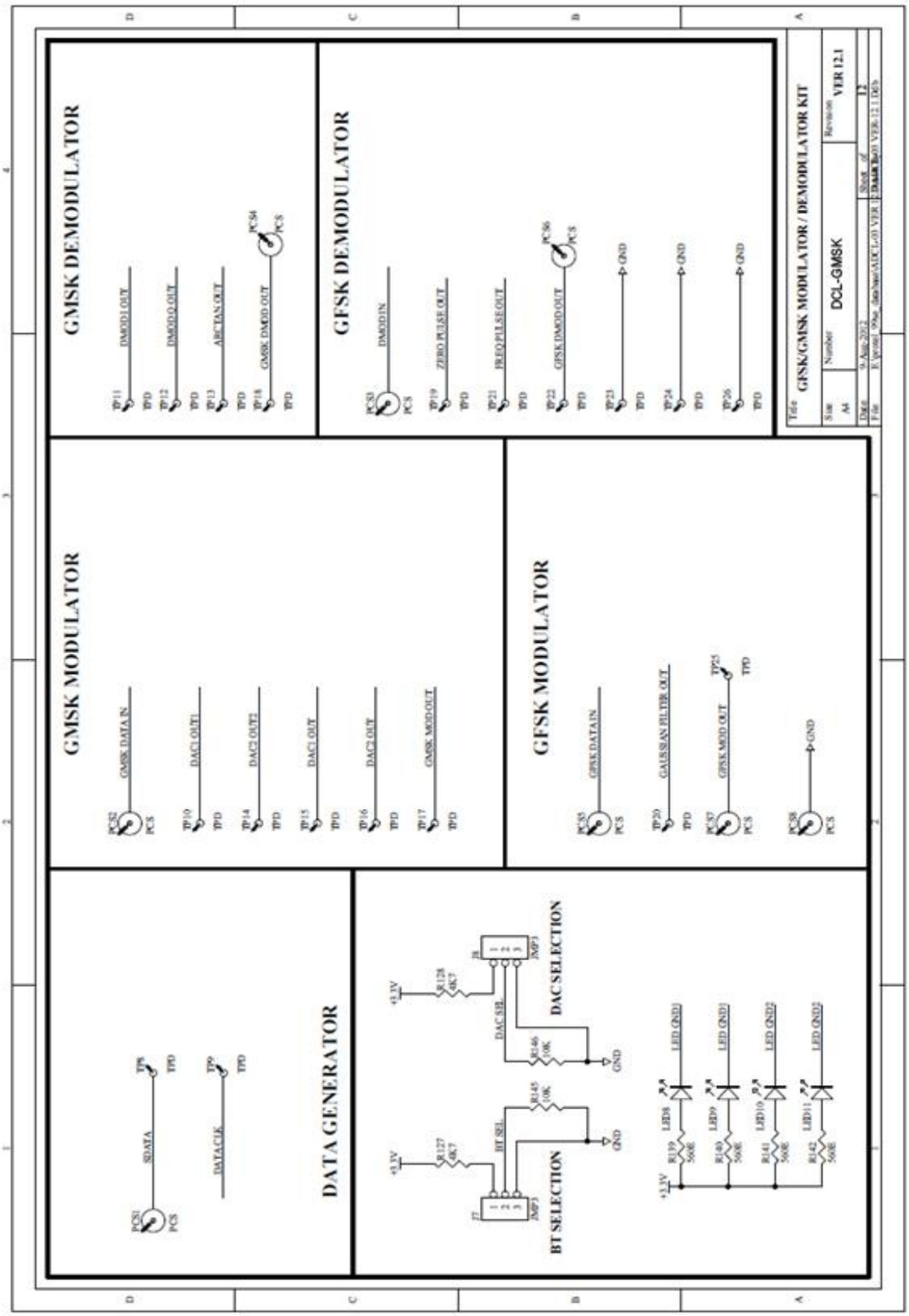
DCL-GMSK
VER 12.1


GMSK
DCL-GMSK
GFSK/GMSK MODULATOR / DEMODULATOR KIT


+3.3V
+2.5
+3.3V
+1.2V
GND
Title
GFSK/GMSK MODULATOR / DEMODULATOR KIT
Size
A4
Number
DCL-GMSK
Revision
VER 12.1
Date
Sheet of
10


+3.3V
+2.5
+3.3V
+1.2V
GND
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1


GMSK DEMODULATOR
DMOD I OUT
DMOD Q OUT
ARCTAN OUT
GMSK DMOD OUT
GFSK DEMODULATOR
DMOD IN
ZERO PULSE OUT
PREQ PULSE OUT
GFSK DMOD OUT
GMSK MODULATOR
GMSK DATA IN
DAC1 OUT1
DAC2 OUT2
DAC1 OUT
DAC2 OUT
GMSK MOD OUT
GFSK MODULATOR
GFSK DATAIN
GAUSSIAN FILTER OUT
GFSK MOD OUT
DATA GENERATOR
SDATA
DATACLK
BT SELECTION
DAC SELECTION
GFSK/GMSK MODULATOR / DEMODULATOR KIT
DCL-GMSK
VER 12.1

EXPERIMENT

NO. 1

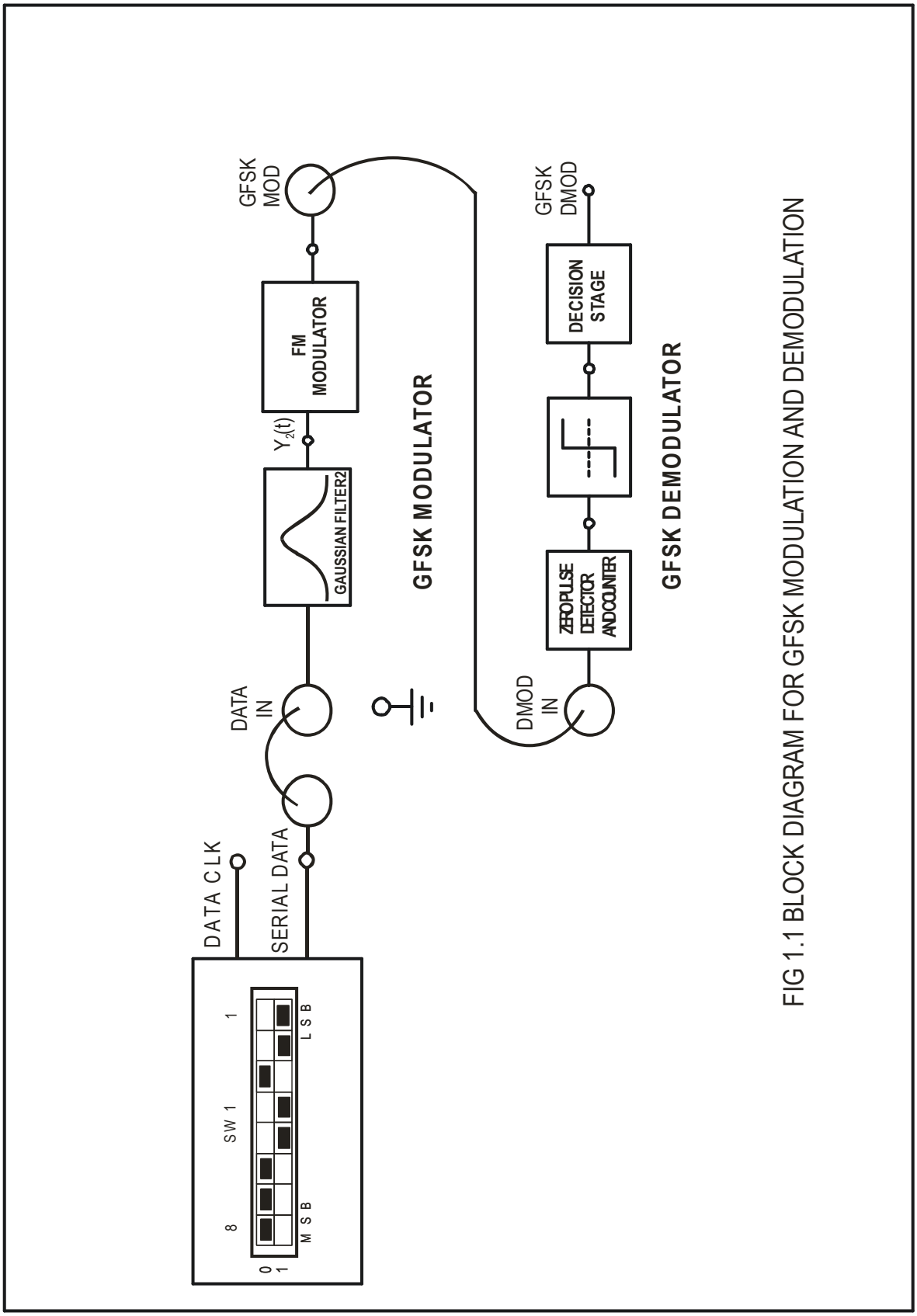


FIG 1.1 BLOCK DIAGRAM FOR GFSK MODULATION AND DEMODULATION

EXPERIMENT NO: 1

NAME:

GFSK MODULATION AND DEMODULATION TECHNIQUE.

OBJECTIVE:

Study of GFSK Modulation and Demodulation.

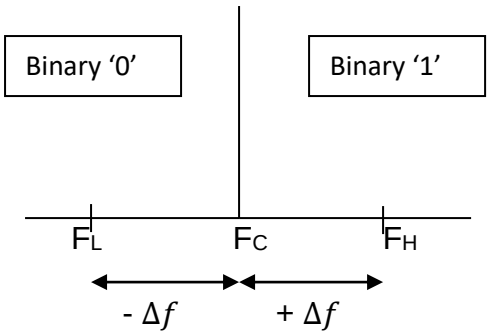
THEORY:

GFSK is a kind of continuous phase frequency shift keying (CPFSK) method having a constant envelope, this allows the use of relatively nonlinear power amplifiers. Also as the waveform produced by the GFSK modulator is continuous in phase even at the edges of the symbols, this system has some containment in frequency, which allows for more channels to be accommodated in a given bandwidth. Also as in GFSK modulation the information is carried in frequency, the additive noise doesn't impact in a direct manner.

A GFSK modulator is composed of an FM modulator with a Gaussian filter applied after the rectangular pulse shaping filter that is used to modulate the phase. The Gaussian filter is used in order to further improve the spectral characteristics of the modulated signals with a tradeoff cost in terms of introducing some inter-symbol interference. GFSK modulators are normally characterized by the product term BT where B is the -3dB bandwidth of the Gaussian filter and T is the symbol duration. For example, the Bluetooth standard is based on GFSK modulation with BT = 0.5.

GFSK MODULATION:

In Gaussian Frequency Shift Keying (GFSK) a binary one is represented by a positive frequency deviation, and a binary zero is a negative frequency deviation.



Where + Δf = positive frequency deviation
- Δf = negative frequency deviation
F_C = center frequency
F_H = high frequency

F_L = low frequency

The binary bit stream is passed through a Gaussian low pass filter that has a BT product of 0.5. Accordingly, for a binary data rate of 256 Kbits/sec the filter's -3.0 dB Bandwidth is 128 kHz.

Filter's -3 dB cutoff = BT product x Bit rate
= 0.5 x 256 e3
= 128 e3
= 128 kHz

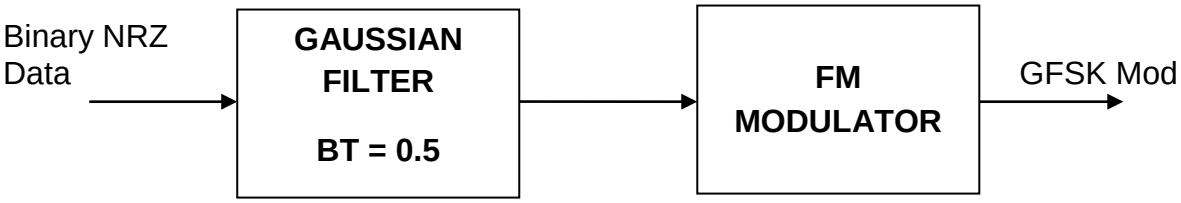


FIG. 1.2 GFSK MODULATOR

A GFSK modulator is same as FSK modulator, except that before the baseband data pulses (-1, 1) go into the FSK modulator, they are passed through a Gaussian filter to make the pulse smoother to limit the transmitter's spectral bandwidth i.e. the bandwidth can be made narrower if the phase is forced to be continuous at the symbol transitions. This implies that the GFSK must look like an FM signal. With GFSK, the frequency climbs and falls slowly to the two frequency extremes. (With
FSK the frequency jumps quickly to the two states.)

In DCL-GMSK, for GFSK modulation techniques, the modulated output shifts between frequencies as per table given.

Y_2 (t)	GFSK MOD
0	2MHz
1	4MHz
0 to 1 OR 1 to 0 transition	In between 2 and 4 MHz

GFSK DEMODULATION:

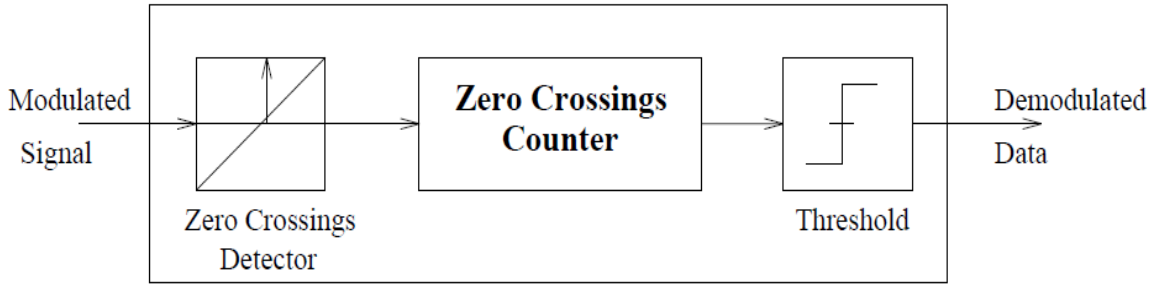


FIG. 1.3 GFSK DEMODULATOR

Zero Crossing Counter: Since the MSB of the input signal gives the zero crossings, counting the number of clock cycles between two rising edges of the MSB will give count between zero crossings which in turn gives a measure of the carrier frequency.

Threshold detector gives logic ‘0’ at output for below threshold value and logic ‘1’ for above threshold value. Threshold value is the half of no. of sampling clock cycles per center frequency.

EQUIPMENTS:

- DCL-GMSK Kit
- Patch Chords
- Power supply
- DSO

PROCEDURE:

1. Refer to the **Fig. 1.1** and carry out the following connections.

2. Connect power supply in proper polarity to the kit **DCL-GMSK** and switch it on.

3. Select a data pattern using switch **SW1** of **DATA GENERATOR BLOCK**.

4. Connect **SERIAL DATA** generated on board to **DATA IN** of **GAUSSIAN FILTER2** of **GFSK MODULATOR** section.

5. Observe the Gaussian shaped pulse at output of **GAUSSIAN FILTER2** block of **GFSK MODULATOR** section.

6. Observe the GFSK modulated signal at output of **GFSK MODULATOR** block of **GFSK MODULATOR** section.

7. For GFSK demodulation connect **GFSK MOD** post to **DMOD IN** post of **GFSK DEMODULATOR** block.
8. Observe various waveforms at respective test points as mentioned below.

OBSERVATION:

Observe the following signal on the oscilloscope for the selected data pattern, using the steps given in the procedure of this experiment.

1. **SERIAL DATA** with respect to **DATA CLK**. **(FIG. 1.4 (a))**
2. **Y_2 (t)** with respect to **DATA IN.** **(FIG. 1.4 (b))**
3. **GFSK MOD** with respect to **Y_2 (t)**. **(FIG. 1.4 (c))**
4. Output of **ZERO PULSE DETECTOR AND COUNTER** at its test point with respect to **GFSK MOD**. **(FIG. 1.4 (d))**
5. Output of **THRESHOLD DETECTOR** at its test point with respect to **GFSK MOD**. **(FIG. 1.4 (e))**
6. **GFSK DMOD** with respect to **SERIAL DATA**. **(FIG. 1.4 (f))**

CAPTURED WAVEFORM

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)

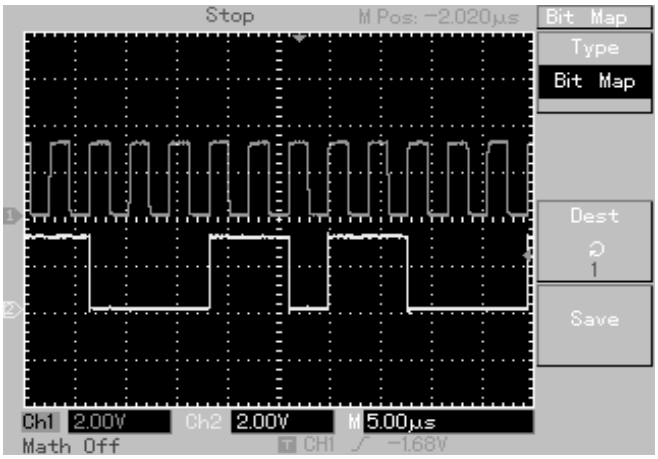


FIG. 1.4 (a)

CH 1: DATA IN & CH 2: Y_2 (t)

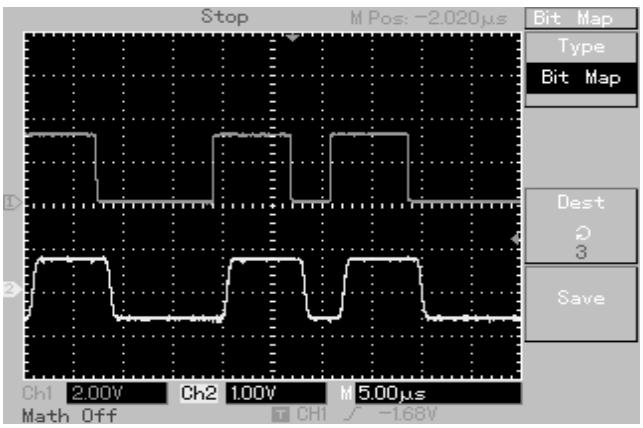


FIG. 1.4 (b)

CH 1: Y_2 (t) & CH 2: GFSK MOD

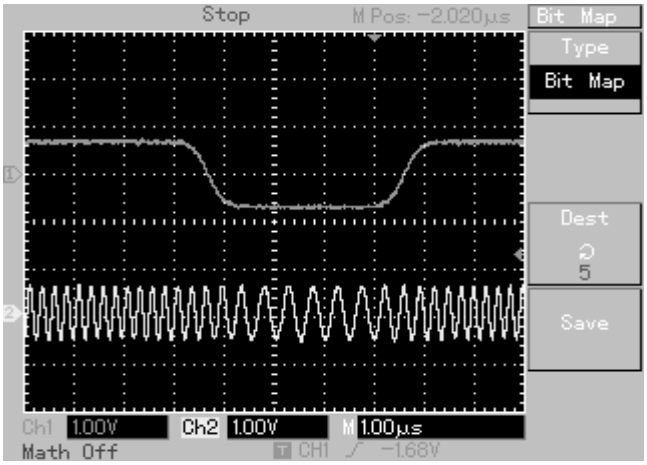


FIG. 1.4 (c)

CH 1: GFSK MOD & CH 2: ZERO PULSE DETECTOR OUT

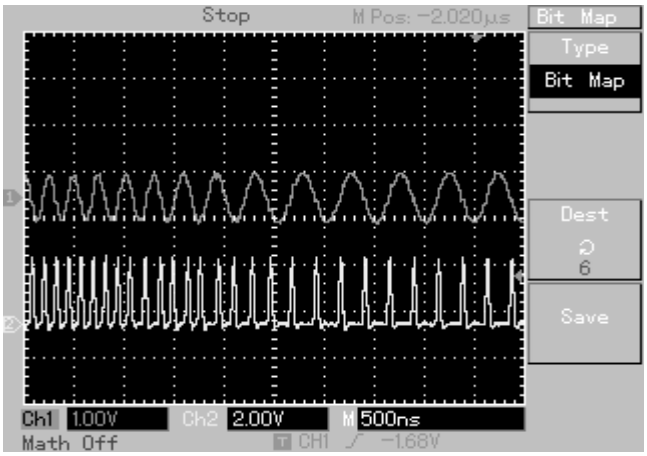


FIG. 1.4 (d)

CH 1: GFSK MOD & CH 2: THRESHOLD DETECTOR

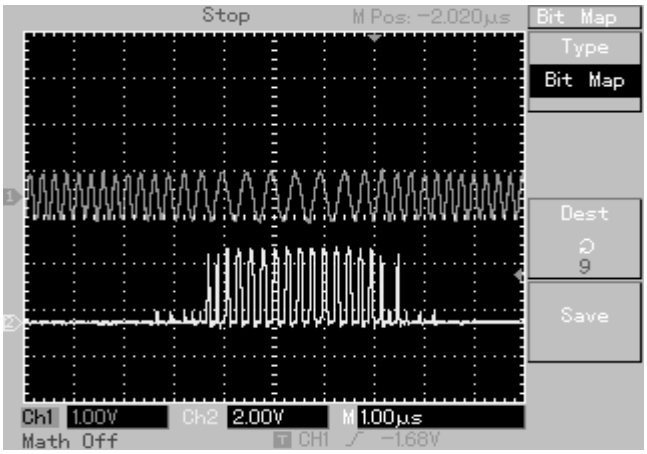


FIG. 1.4 (e)

CH 1: SERIAL DATA & CH 2: GFSK DMOD

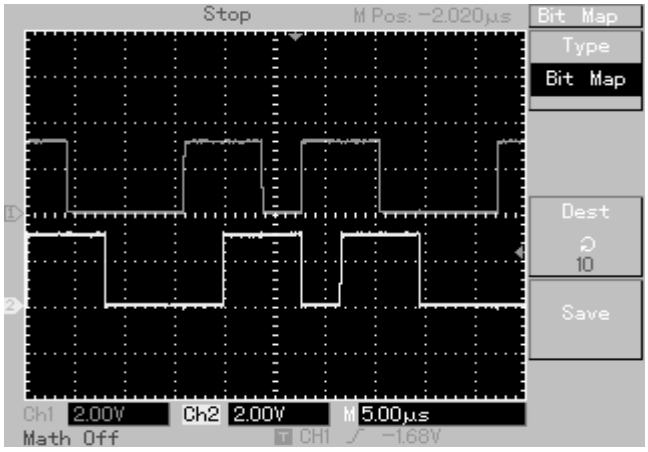


FIG. 1.4 (f)

CONCLUSION:

In this experiment we have studied GFSK modulation and demodulation technique. We have observed that there are some frequency present in between mark and space at the ‘one’ to ‘zero’ and ‘zero’ to ‘one’ transition.

EXPERIMENT

NO. 2

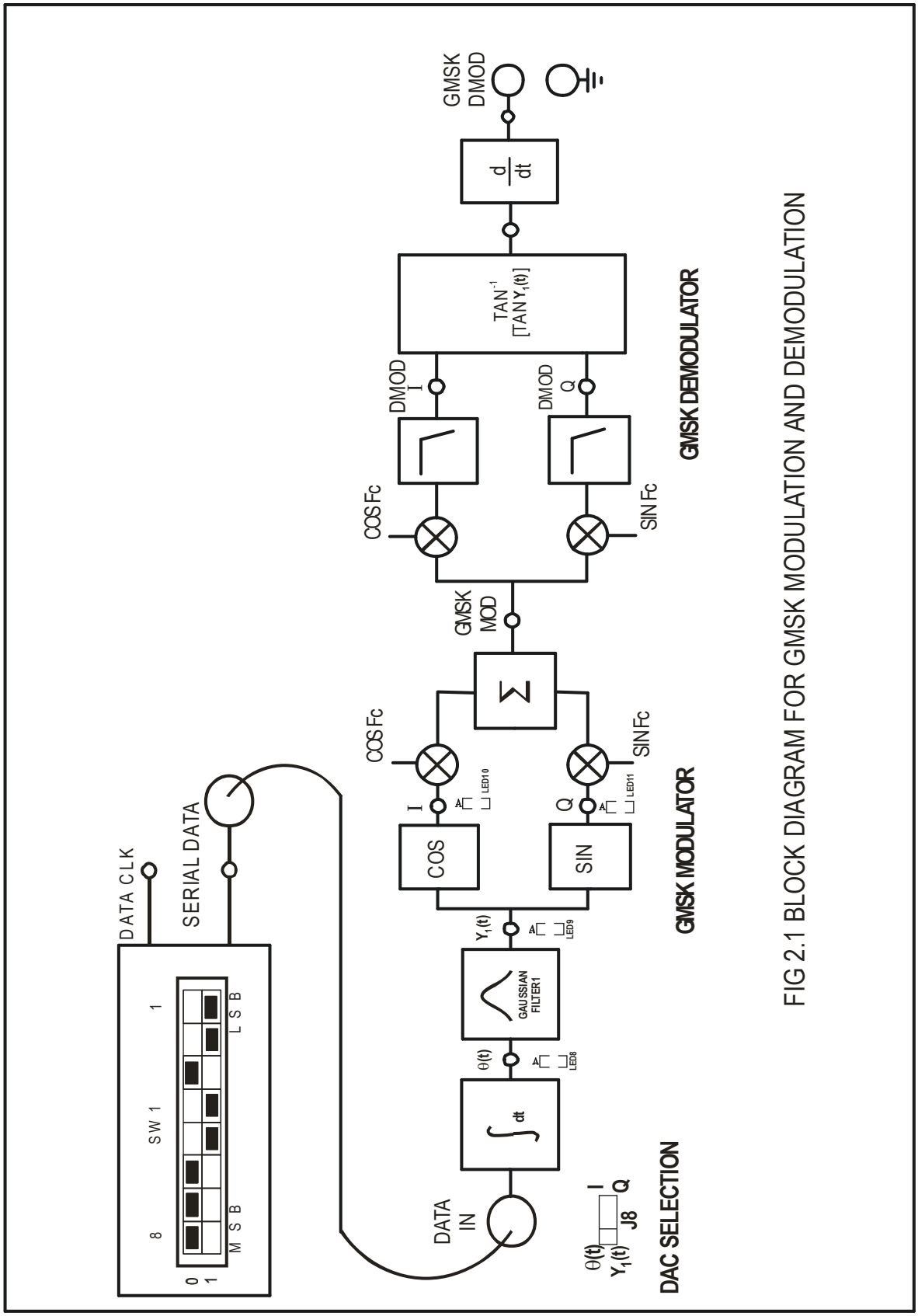


FIG 2.1 BLOCK DIAGRAM FOR GMSK MODULATION AND DEMODULATION

EXPERIMENT NO: 2

NAME:

GMSK MODULATION AND DEMODULATION TECHNIQUES.

OBJECTIVE:

Study of GMSK Modulation and Demodulation.

THEORY:

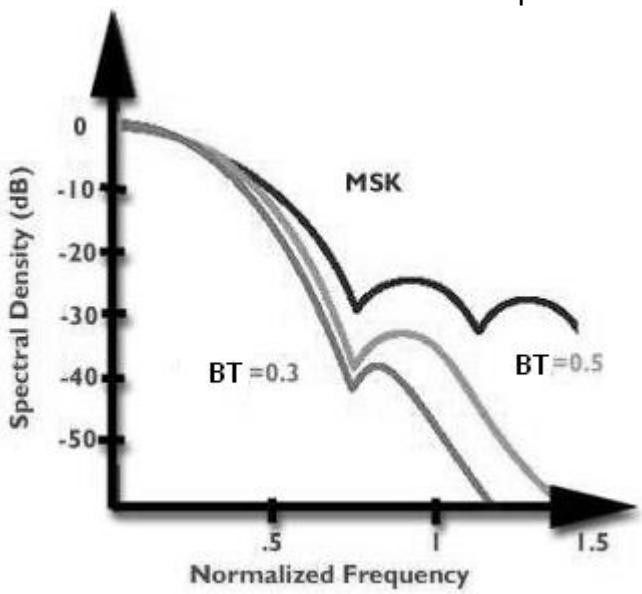
GMSK finds a wide range of usage due to its spectral efficiency. As GMSK is a type of MSK scheme, it has a modulation index of 0.5. It differs from the MSK in the aspect of the usage of the filter. GMSK is a result of the attempts to improve the MSK power spectrum.

The Gaussian filter concentrates the energy on a desired band for low out of band power characteristic. Widely known advantages of GMSK, that are narrow bandwidth and constant envelope modulation, make the GMSK suitable for both coherent and incoherent detection. Due to the constant envelope scheme, that is a property of GMSK, makes it less susceptible to fading environments than amplitude modulation and it requires inexpensive class-C amplifiers to be utilized for this scheme.

GMSK has a low out of band power characteristics and constant envelope which makes it a desirable choice for usage in a wireless mobile communications. The effect of the filter brings out the suppression of the out of band power by its sharp cut-off property.

Frequency Response:

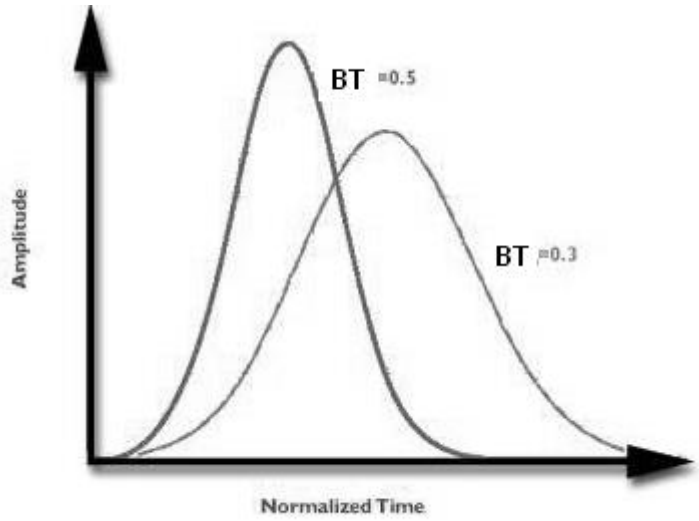
The principle parameter in designing an appropriate Gaussian filter is the time bandwidth product BT. Please see the following figure for the frequency response of different Gaussian filters. Note that MSK has a time-bandwidth product of infinity.



As can be seen from above, GMSKs power spectrum drops much quicker than MSK's. Furthermore, as BT is decreased, the roll-off is much quicker.

Time-Domain Response:

Since lower time-bandwidth products produce a faster power-spectrum roll-off, why not has a very small time-bandwidth product. It happens that with lower time-bandwidth products the pulse is spread over a longer time, which can cause inter symbol interference.



Therefore as a compromise between spectral efficiency and time-domain performance, an intermediate time-bandwidth product must be chosen.

Advantages of GMSK modulation

There are several advantages to the use of GMSK modulation for a radio communications system. One is obviously the improved spectral efficiency when compared to other phase shift keyed modes.

A further advantage of GMSK is that it can be amplified by a non-linear amplifier and remain undistorted. This is because there are no elements of the signal that are carried as amplitude variations. This advantage is of particular importance when using small portable transmitters, such as those required by cellular technology. Non-linear amplifiers are more efficient in terms of the DC power input from the power rails that they convert into a radio frequency signal. This means that the power consumption for a given output is much less, and this results in lower levels of battery consumption; a very important factor for cell phones.

A further advantage of GMSK modulation again arises from the fact that none of the information is carried as amplitude variations. This means that is immune to amplitude variations and therefore more resilient to noise, than some other forms of modulation, because most noise is mainly amplitude based.

GMSK MODULATOR:

In DCL-GMSK for GMSK, Gaussian low pass filter has a BT product of 0.3. Accordingly, for a binary data rate of 256 Kbits/sec the filter's -3.0 dB Bandwidth is 76.8 kHz.

Filter's -3 dB cutoff = BT product x Bit rate
= 0.3 x 256 e3
= 76.8 e3
= 76.8 kHz

Modulation index is 0.5. The modulation index (μ) is the frequency deviation divided by bit rate. Therefore:

Frequency deviation = μ x bit rate
= 0.5 x 256 e3
= 0.128 e6
= 128 kHz

With a frequency deviation of 128 kHz, the RF carrier for a binary 0 will be offset -64 kHz, and a binary 1 will produce a +64 kHz offset.

Center Frequency = Fc = 512 KHz.

Frequency of the high frequency carrier = 512 KHz + 64 KHz = 576 KHz.

Frequency of the low frequency carrier = 512 KHz - 64 KHz = 448 KHz.

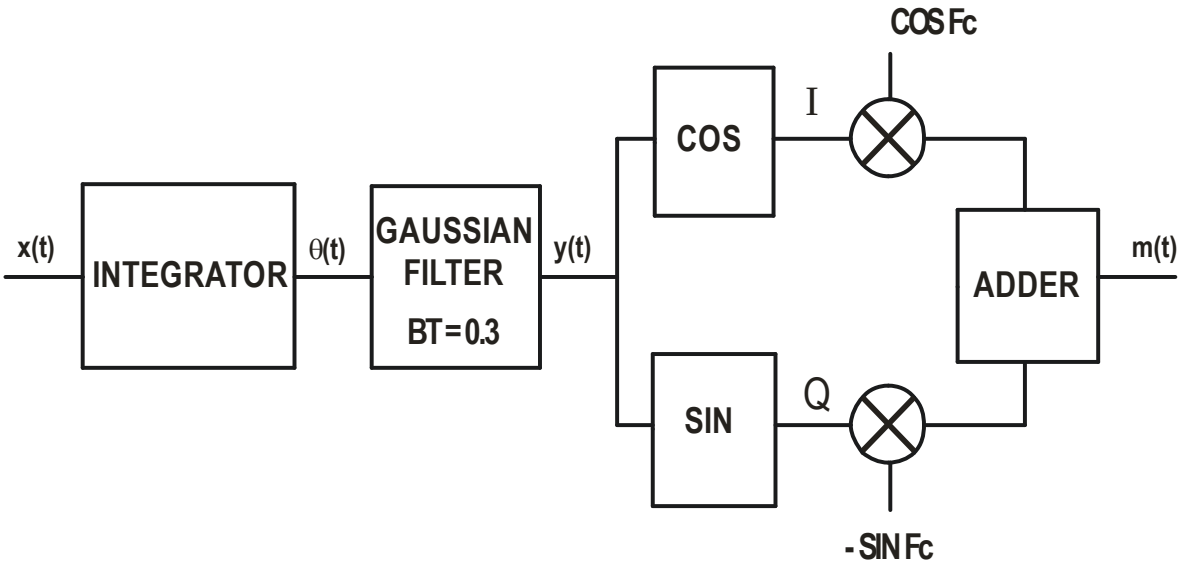


Fig. 2.2 GMSK Modulator Block Diagram

- Input NRZ sequence x(t) is fed to the Integrator.
- The Integrator integrates the NRZ sequence to give the phase plot.

$$\theta(t) = \int x(t)\, dt$$

- To smooth out the phase transition, output of Integrator is passed through the Gaussian Filter which gives the continuous phase change.

$$y(t) = h(t) * \theta(t)$$

- The I and Q components are obtained from Cos LUT and Sin LUT, using the Gaussian Filter output as an address. This I and Q are the quadrature components of the baseband GMSK equivalent signals.

$$I(t) = \cos [y(t)]$$
$$Q(t) = \sin [y(t)]$$

- By multiplying the I and Q components with the corresponding cos(Fc) and –sin(Fc) carriers the modulate I and Q signals are obtained .

- By adding these two modulated signals GMSK modulated signal is obtained.

$$m(t) = I(t)\cos(Fc) – Q(t) \sin(Fc)$$

GMSK DEMODULATOR:

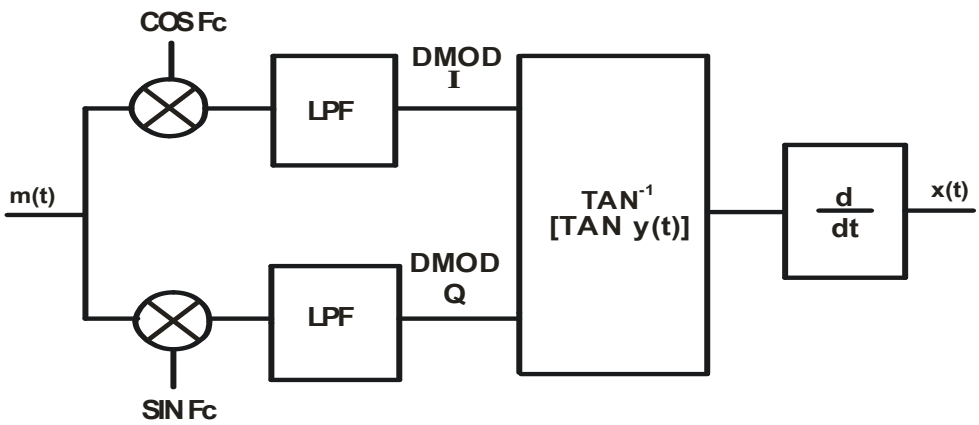


Fig. 2.3 GMSK Demodulator Block Diagram

- Modulated GMSK signal m(t) is fed to the GMSK DEMODULATOR.
- Multipliers at demodulator remove the phase from the modulated signal.
- Low Pass Filter by pass the high frequency component and follow the shape of its input. The demodulated I and Q are obtained at the output of Low Pass Filters.
- TAN^{-1} block is used to recover the phase plot.

i.e. $\tan^{-1}[\tan y(t)] = y(t)$

where $\tan[y(t)] = \frac{\text{dmod Q}}{\text{dmod I}}$

$$= \frac{\sin[y(t)]}{\cos[y(t)]}$$

- Derivator is used to recover the input NRZ signal.

$$x(t) = \frac{d\, y(t)}{dt}$$

EQUIPMENTS:

- DCL-GMSK Kit.
- Patch Chords.
- Power supply.
- DSO.

PROCEDURE

1. Refer to the **Fig. 2.1** and carry out the following connections.

2. Connect power supply in proper polarity to the kit **DCL-GMSK** and switch it on.

3. Select a data pattern “00011011” using switch **SW1** of **DATA GENERATOR** block.

4. Connect **SERIAL DATA** generated on board to **DATA IN** of **INTEGRATOR** of **GMSK MODULATOR** section.

5. Select a **DAC selection** jumper (**J8**) position as shown in **Fig. 2.1** to observe the respective output of the **GMSK MODULATOR** section. By the setting of jumper (**J8**) the respective LEDs glow. When **LED 8** and **LED 9** glow then observe the output at **θ (t)** and **Y_1 (t)** test points. When **LED 10** and **LED 11** glow then observe the output at **I** and **Q** test points.

6. Observe the GMSK modulated signal at **GMSK MOD** test point of **GMSK MODULATOR** section.

7. Output of **GMSK MODULATOR** is connected to **GMSK DEMODULATOR** internally. Observe the signal of various block of **GMSK DEMODULATOR** at their respective test points.

8. Observe various waveforms at respective test points as mentioned below.

OBSERVATION

Observe the following signal on the oscilloscope for the selected data pattern, using the steps given in the procedure of this experiment.

1. **SERIAL DATA** with respect to **DATA CLK**. **(FIG. 2.4 (a))**
2. Output of **INTEGRATOR** at post **θ (t) with** respect to **SERIAL DATA.** **(FIG. 2.4 (b))**
3. Output of **GAUSSIAN FILTER1** at post **Y_1 (t)** with respect to **SERIAL DATA.** (**FIG. 2.4 (c))**
4. **Y_1 (t) with** respect to **θ (t)**. **(FIG. 2.4 (d))**

Note: - Integrator is implemented using up-down counter and it is working on 4MHz clock. When SERIAL DATA is logic ‘1’ it is incremented and for logic ‘0’ it is decremented. For the Data pattern “00011011” Integrator works smoothly. For data pattern other than this there will be possibility of overflow i.e. Integrator output gives sudden transition, due to which glitches will be observed. (Therefore do the whole experiment for data pattern “00011011” by observing each signal at its respective test point.)

5. Output of **COS** block at post **I** and output of **SIN** block at post **Q** **(FIG. 2.4 (e))**
6. Output of **ADDER** at post **GMSK MOD** with respect to **SERIAL DATA.** **(FIG. 2.4 (f))**
7. Output of **LPF** block of **GMSK DEMODULATOR** section at post **DMOD I** with respect to Output of **COS** block at post **I.** **(FIG. 2.4 (g))**
8. Output of **LPF** block of **GMSK DEMODULATOR** section at post **DMOD Q** with respect to Output of **SIN** block at post **Q.** **(FIG. 2.4 (h))**
9. Output of **TAN^{-1}** at its test point with respect to Output of **GAUSSIAN FILTER1** at post **Y_1 (t).** **(FIG. 2.4 (i))**
9. Output of **DERIVATOR** at post **GMSK DMOD** with respect to **SERIAL DATA.** **(FIG. 2.4 (j))**

CAPTURED WAVEFORM:

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)

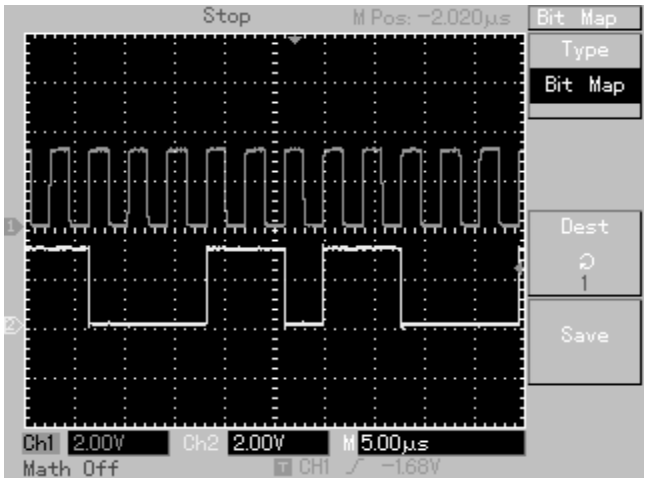


FIG. 2.4 (a)

CH 1: SERIAL DATA & CH 2: θ (t)

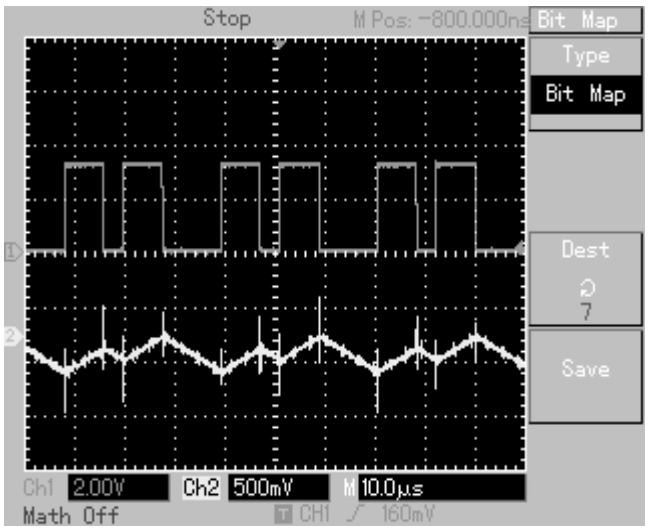


FIG. 2.4 (b)

CH 1: SERIAL DATA & CH 2: Y_1 (t)

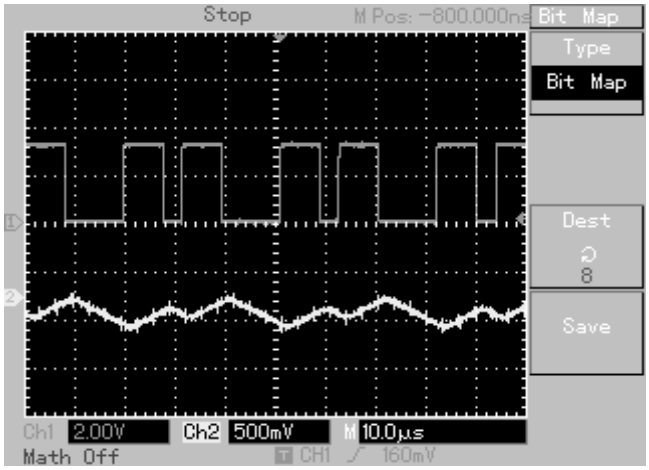


FIG. 2.4 (c)

CH 1: θ (t) & CH 2: Y_1 (t)

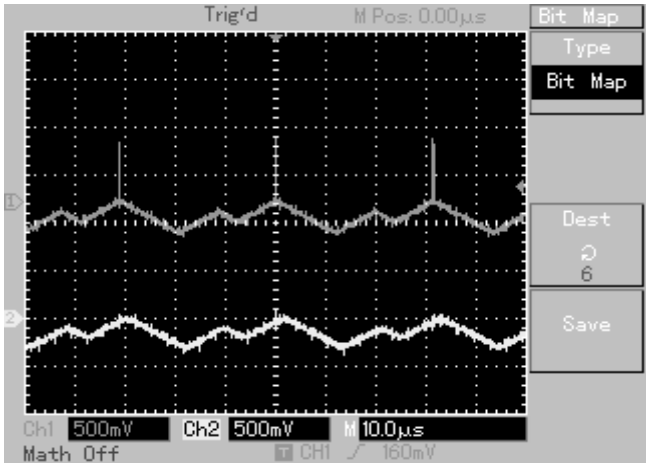


FIG. 2.4 (d)

CH 1: I & CH 2: Q

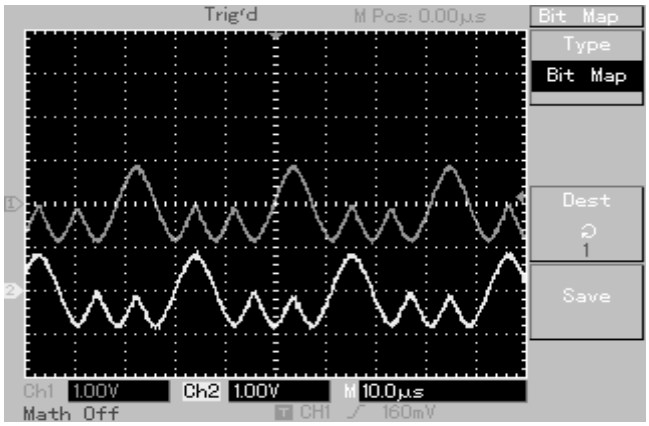


FIG. 2.4 (e)

CH 1: SERIAL DATA & CH 2: GMSK MOD

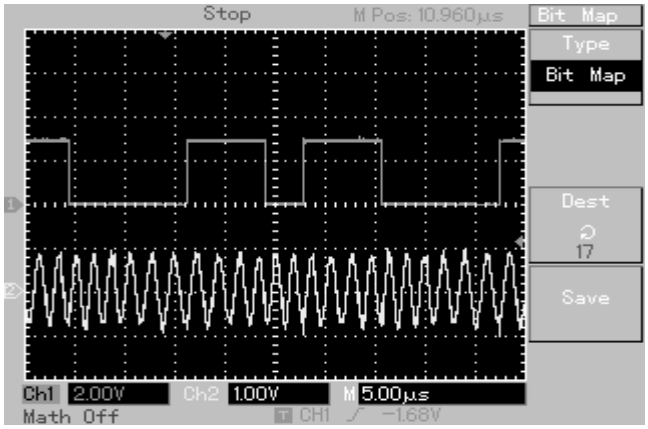


FIG. 2.4 (f)

CH 1: I & CH 2: DMOD I

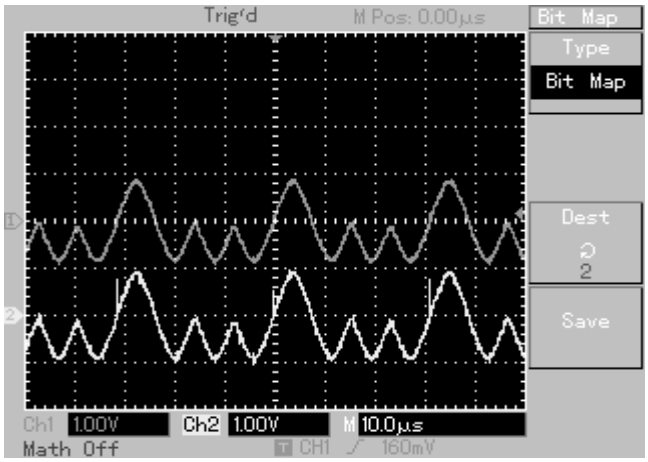


FIG. 2.4 (g)

CH 1: Q & CH 2: DMOD Q

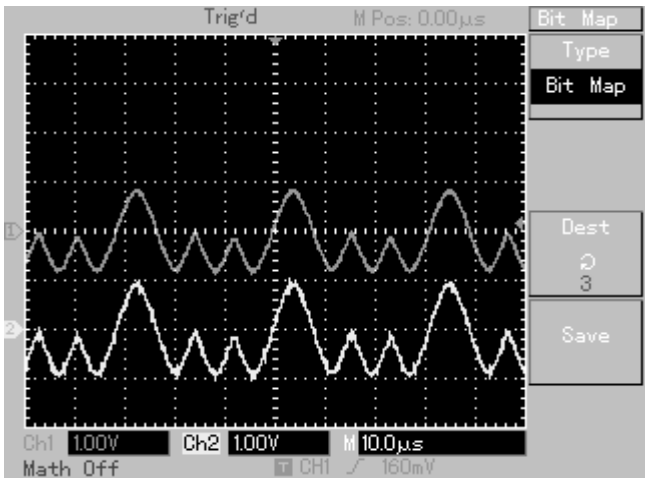


FIG. 2.4 (h)

CH 1: Y_1 (t) & CH 2: OUTPUT OF TAN^{-1} BLOCK

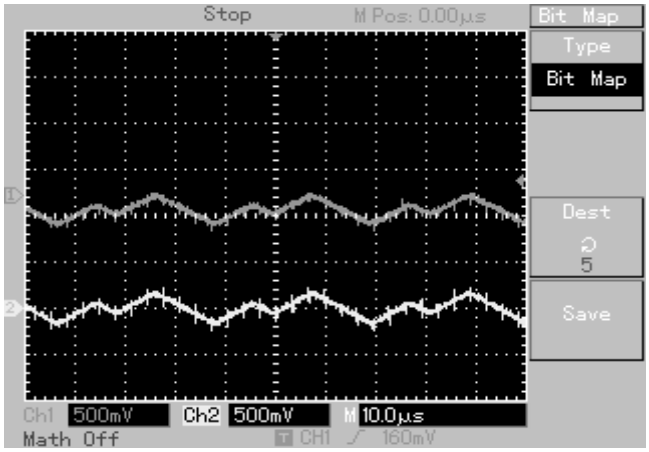


FIG. 2.4 (i)

CH 1: SERIAL DATA & CH 2: GMSK DMOD

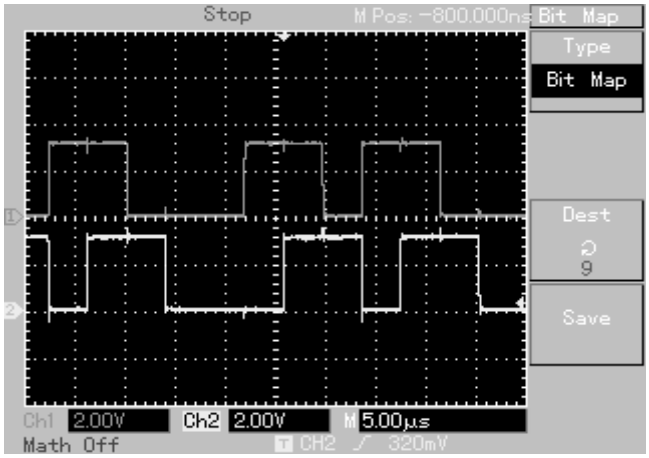


FIG. 2.4 (j)

CONCLUSION:

In this experiment we have studied GMSK modulation and demodulation technique using I&Q modulation whose modulation index is 0.5 and observed various waveforms.